

Monitoring ML services

MLOPS DEPLOYMENT AND LIFE CYCLING



Nemanja Radojkovic
Senior Machine Learning Engineer

Maintaining quality

- Paying customers == expectations of quality
- Quality assurance starts with quality control
- Monitoring

Performance indicators

Fundamental health indicators

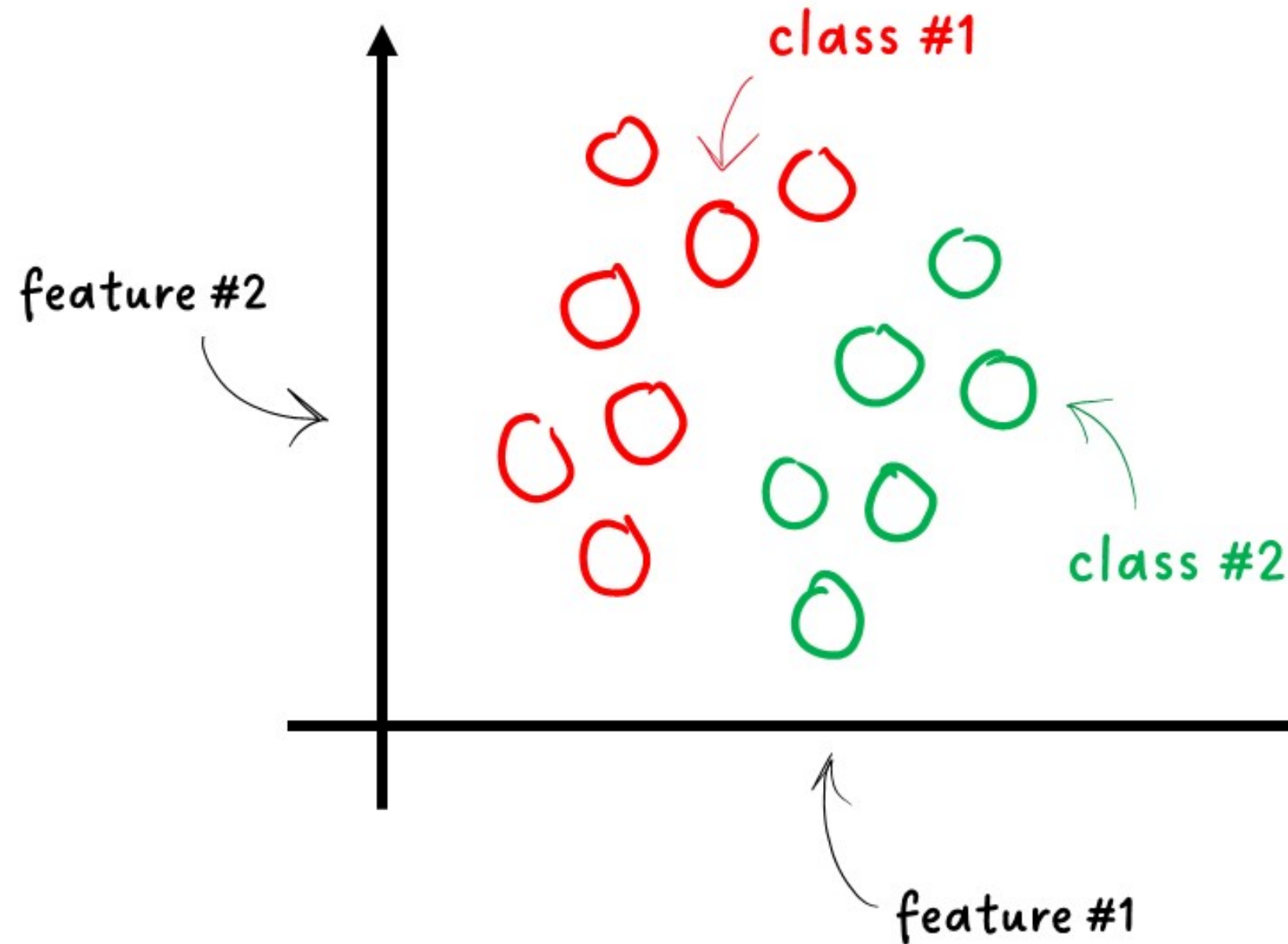
- Service up and running?
- Number of requests in time?
- Latency distribution?

Ultimate quality metric

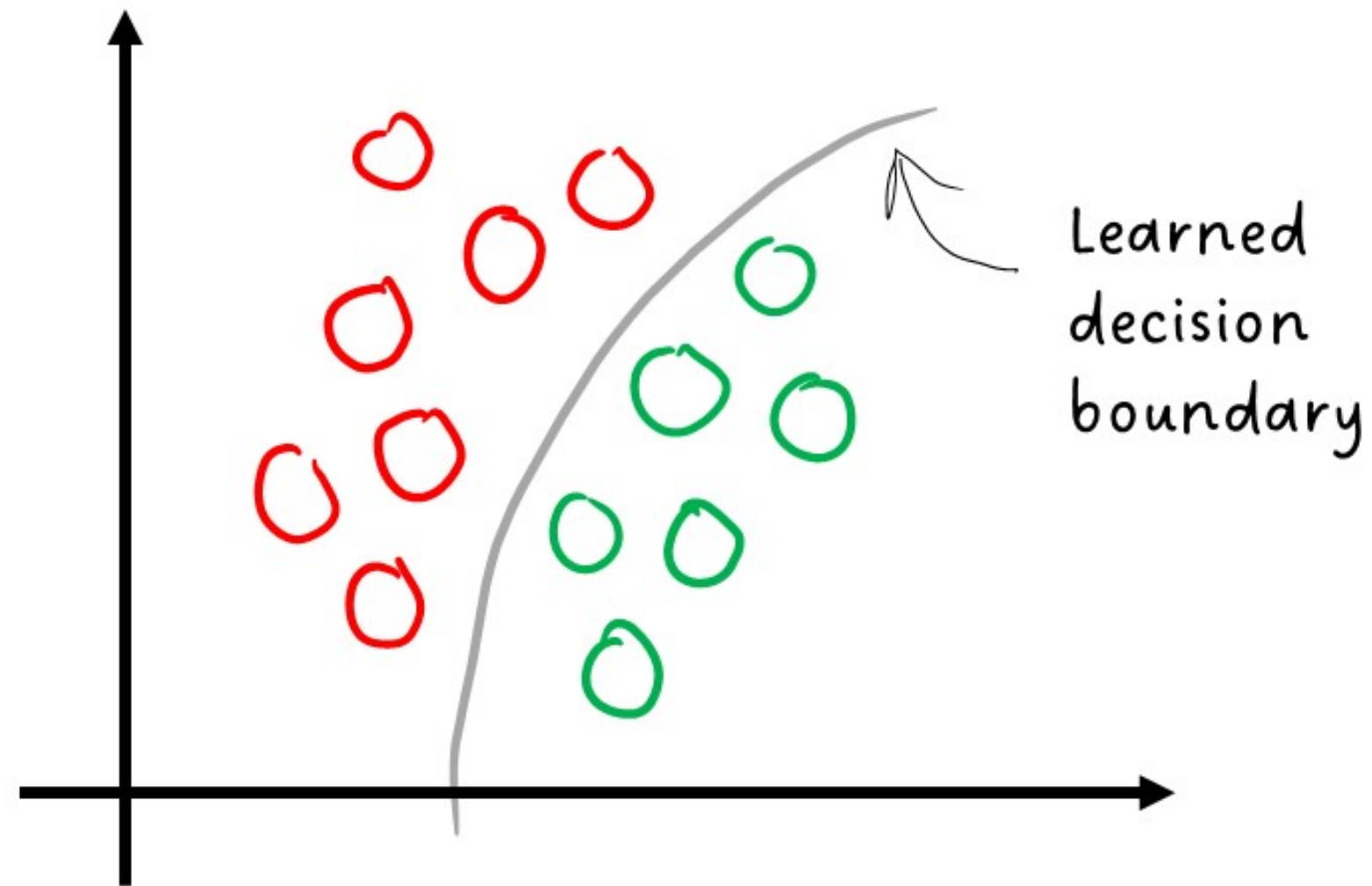
- Predictive performance

How do ML models deteriorate?

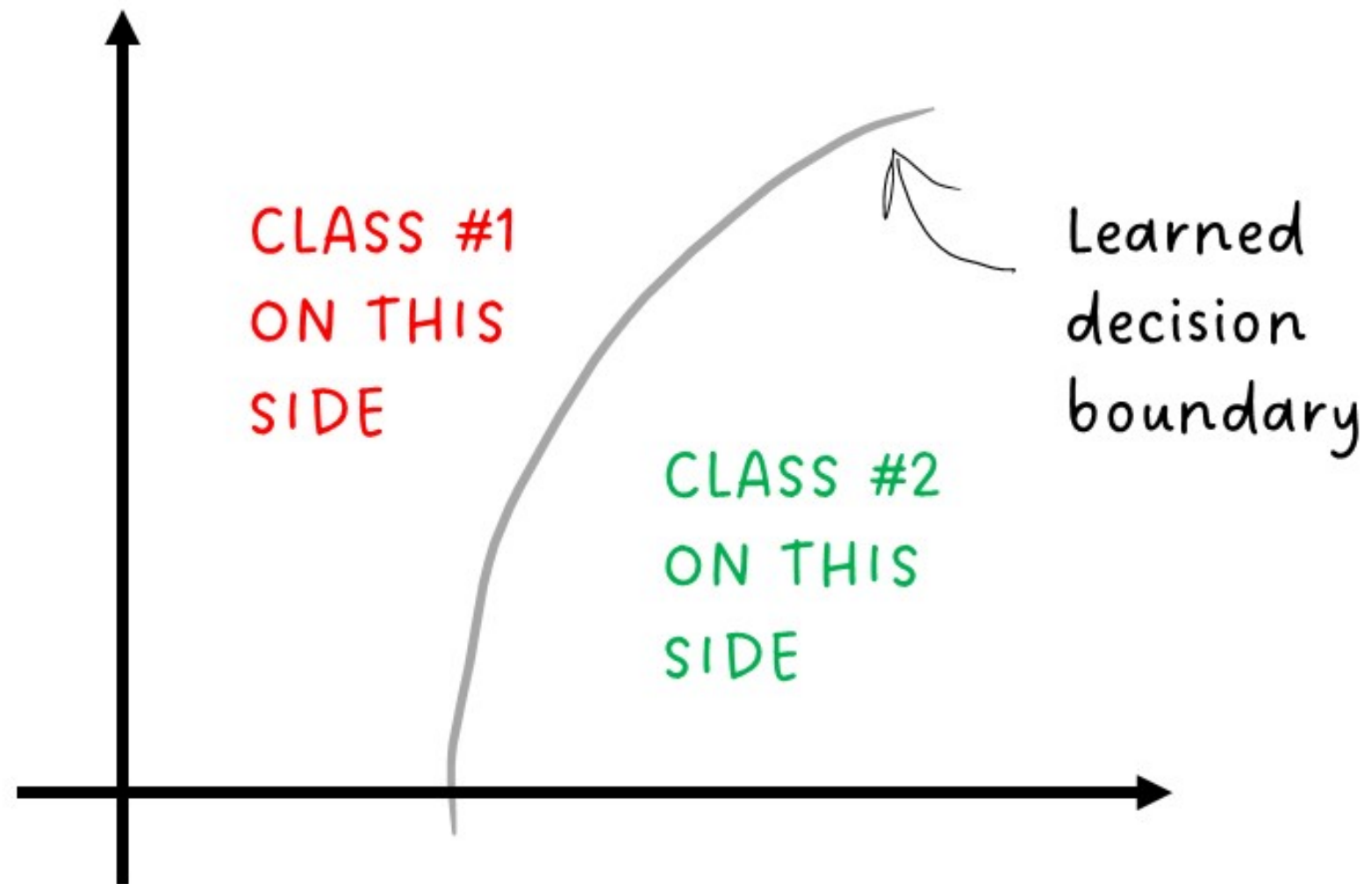
Features and labels at training time



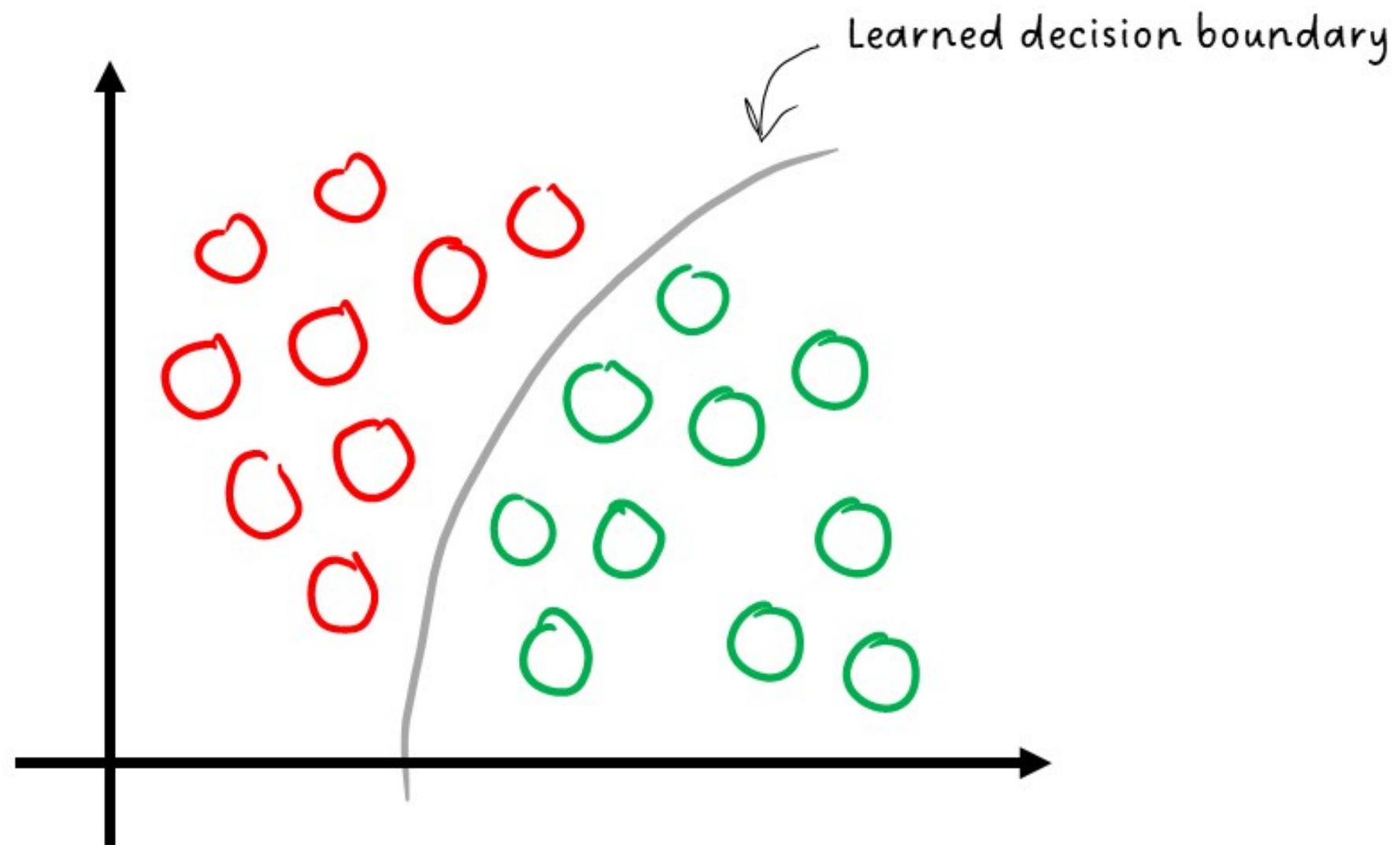
Features and labels at training time



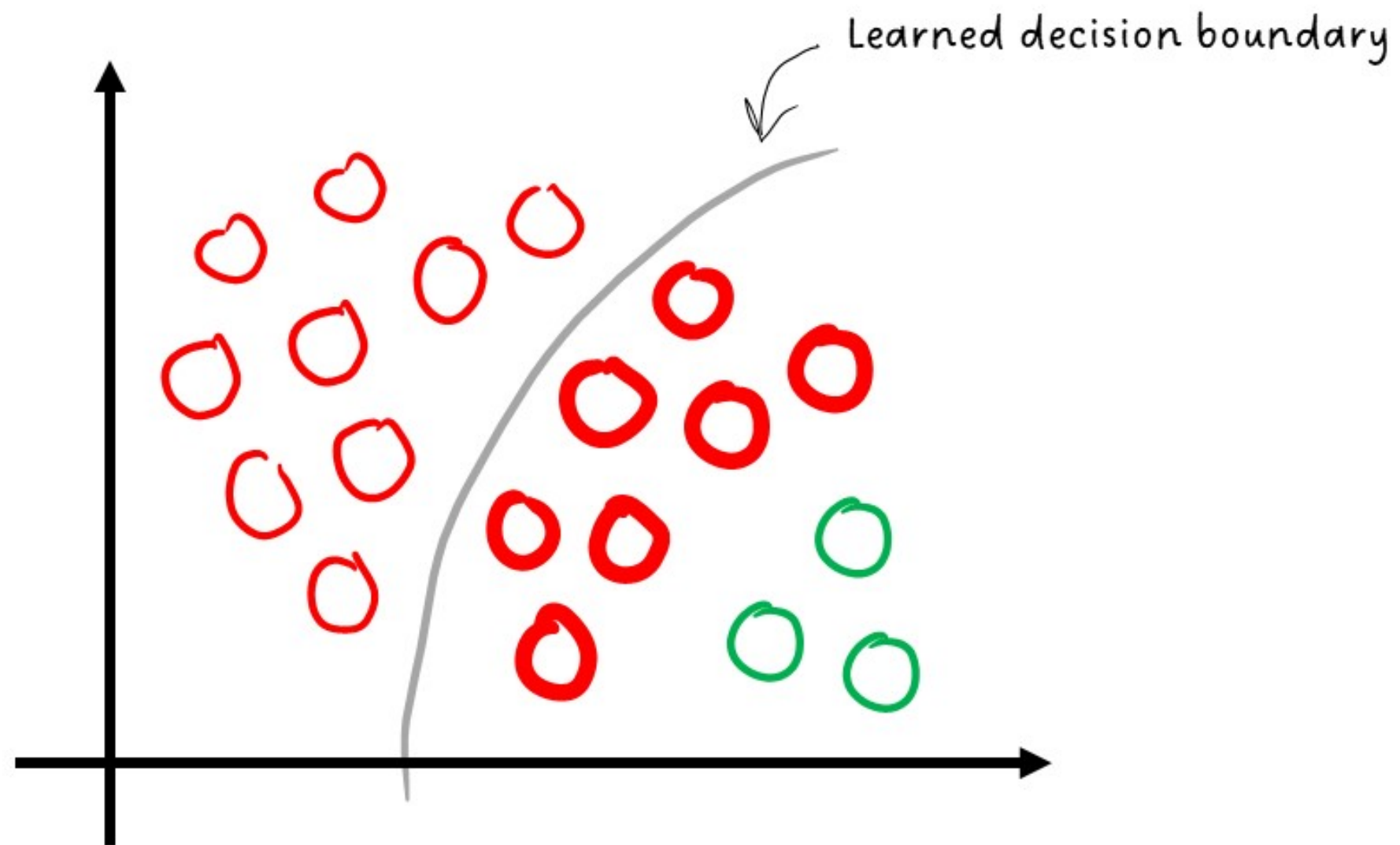
Trained model



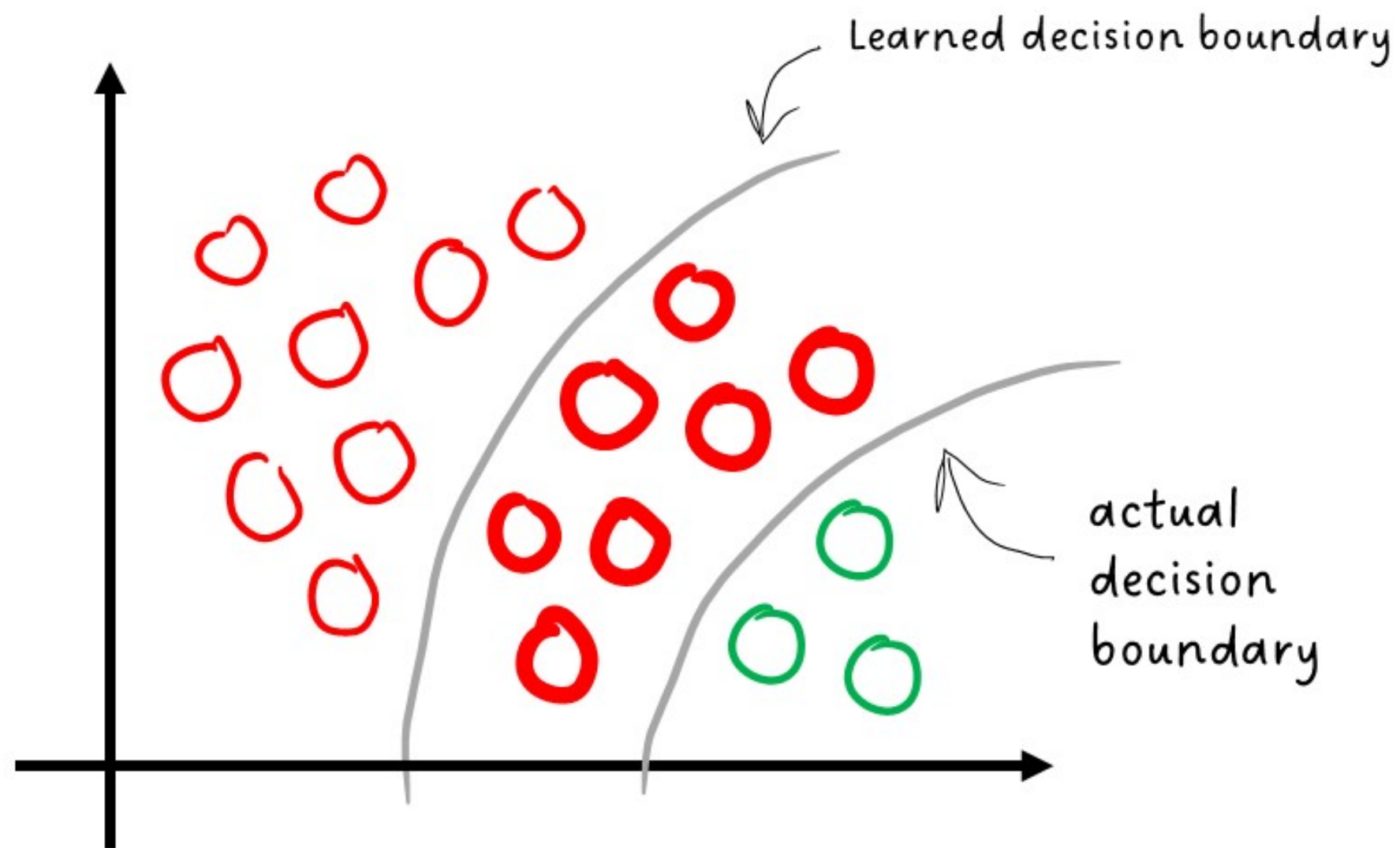
AFTER A WHILE...



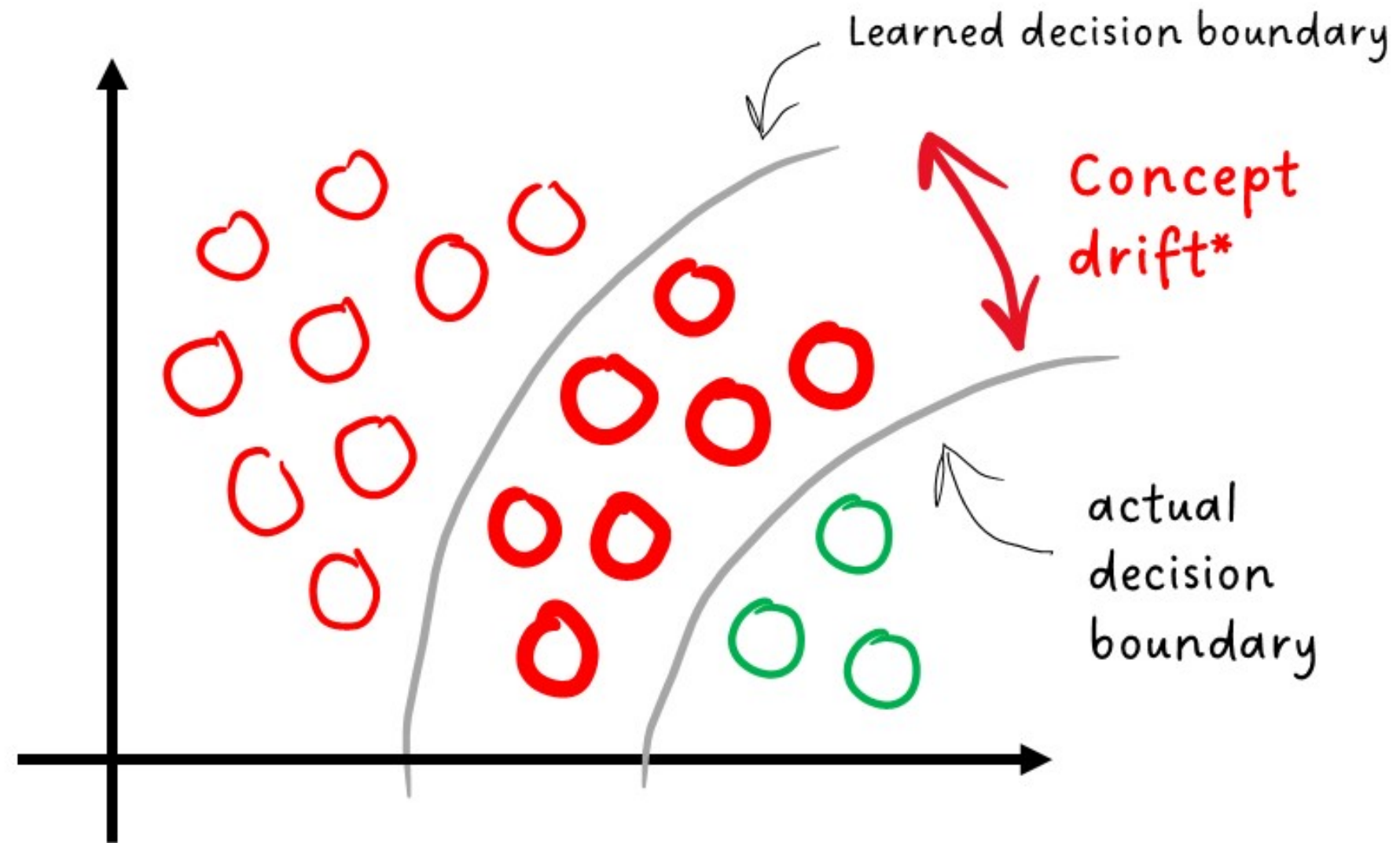
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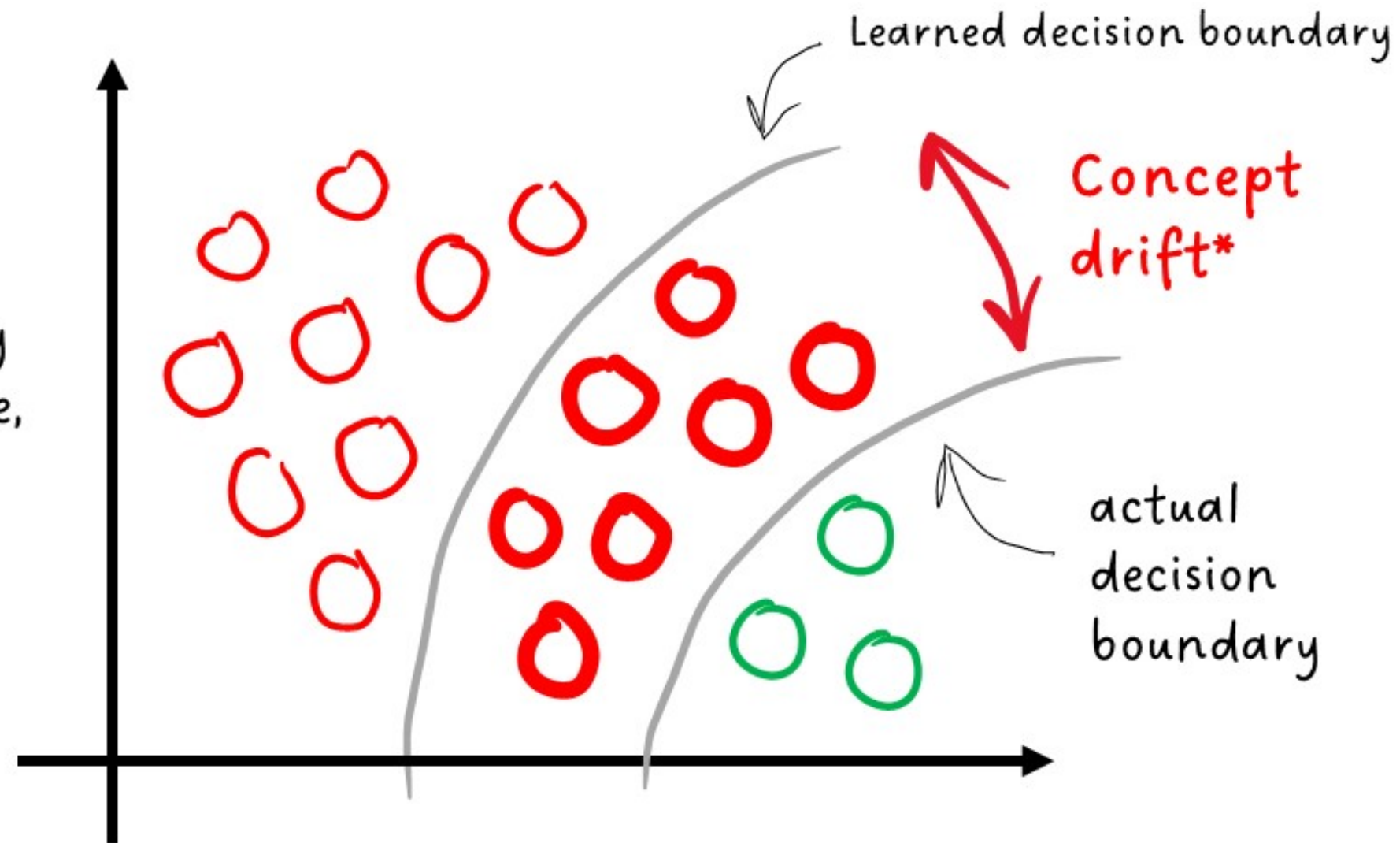
AFTER A WHILE...



*concept drift == significant change in the actual relationship between the input and output features

AFTER A WHILE...

The model
"deteriorates" by
staying the same,
while the reality
changes!



*concept drift == significant change in the actual relationship between the input and output features

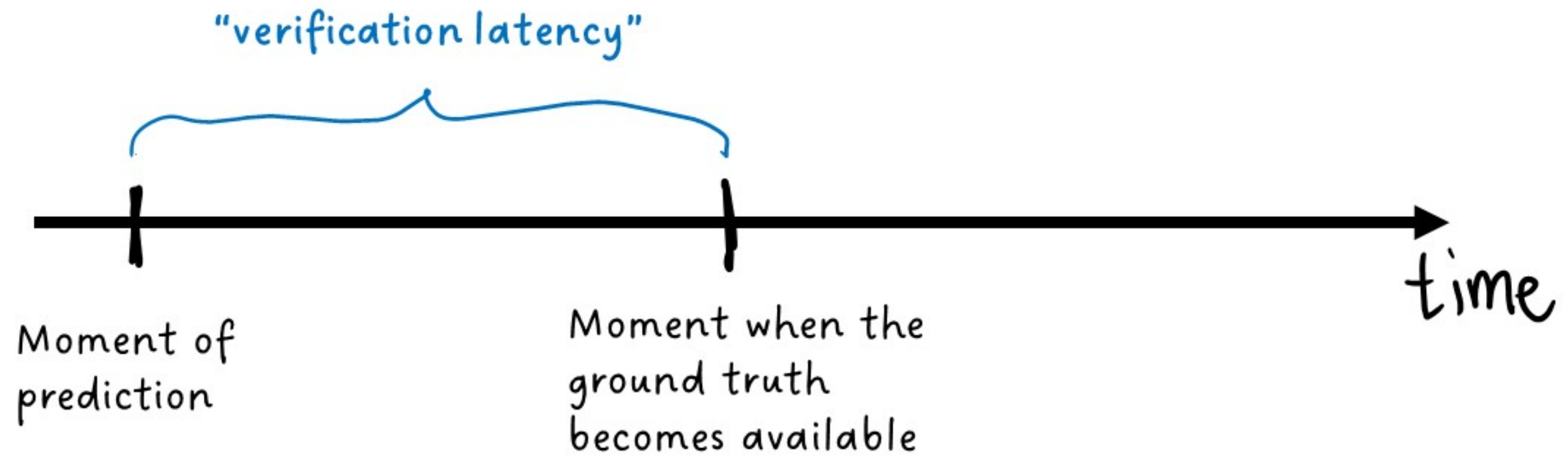
How to detect
concept drift?

How to detect concept drift?



```
IF prediction != ground_truth  
MORE_OFTEN_THAN_EXPECTED  
THEN WARNING("CONCEPT DRIFT!")
```

The catch:



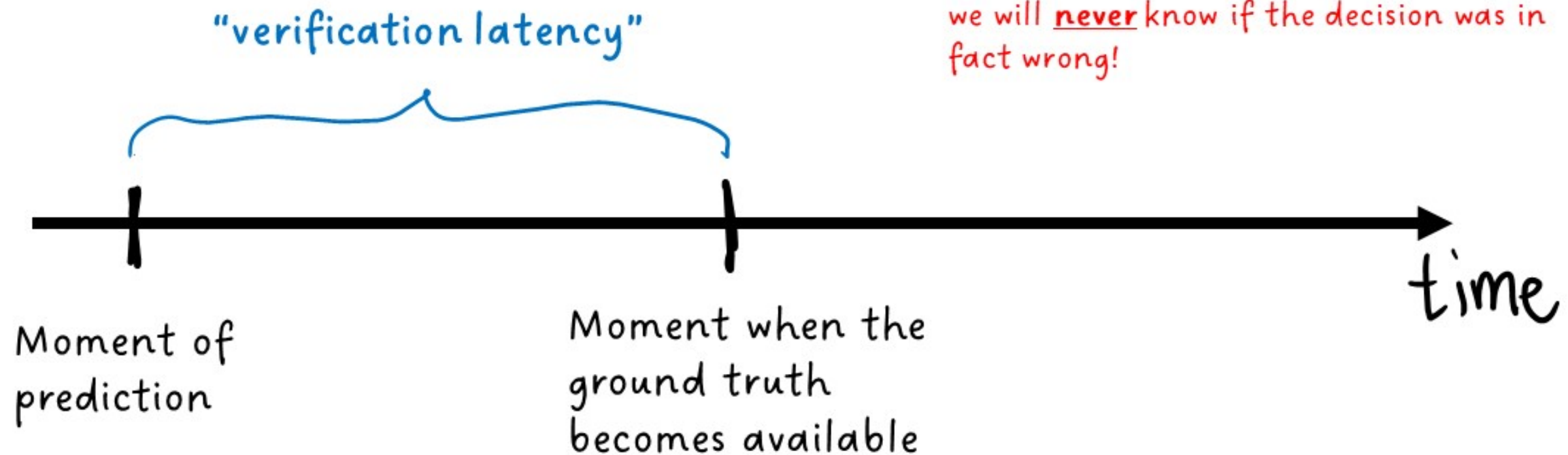
The catch:

Stock prediction: < 1 second

Fraud detection: ~ months

Credit scoring: ∞

If a loan is denied based on our model, we will never know if the decision was in fact wrong!



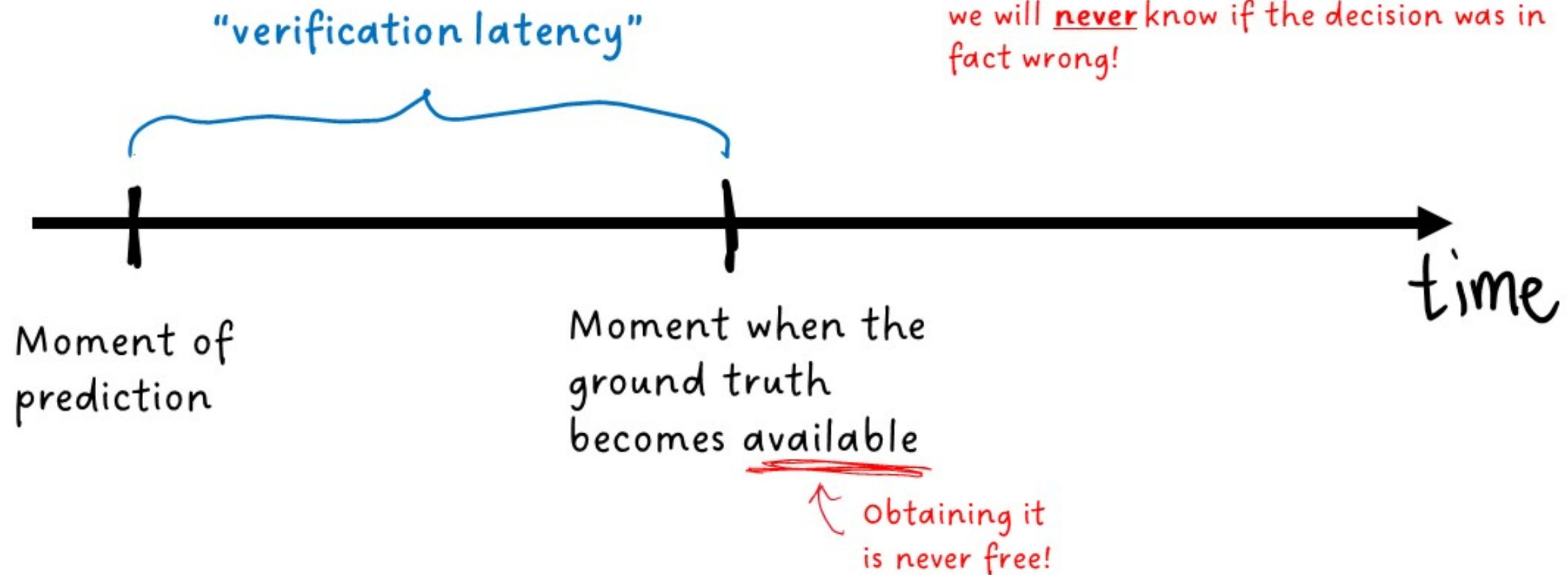
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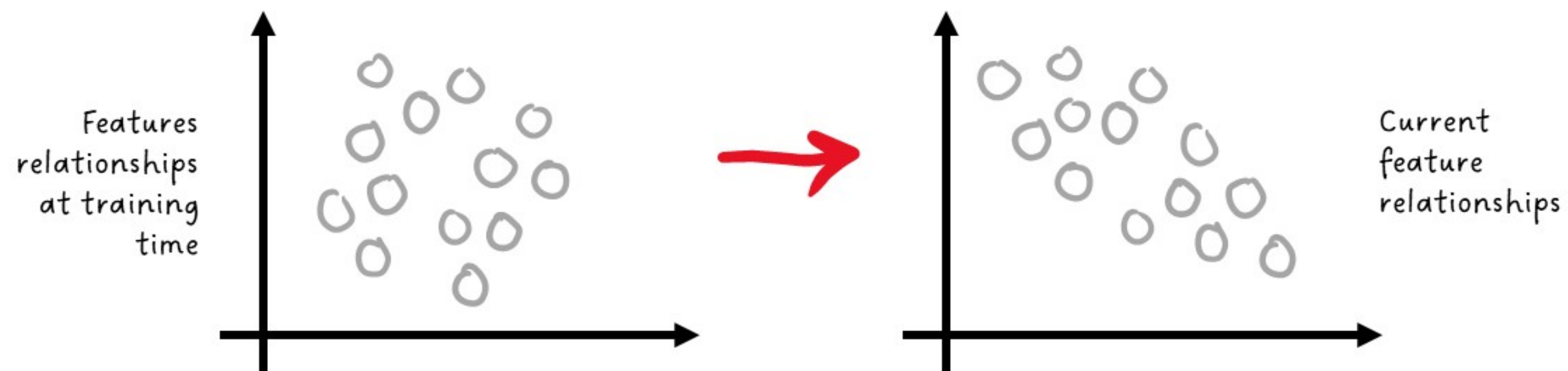
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Indirect approach:

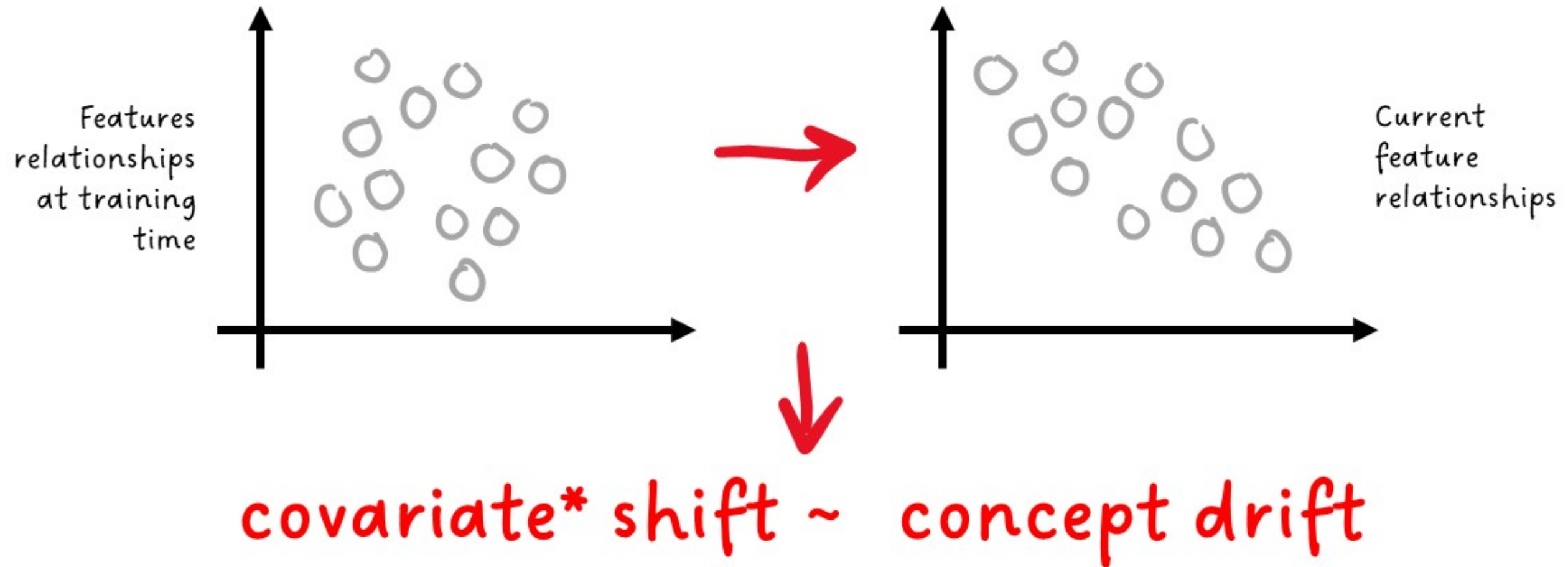
Monitor what you have: input features!

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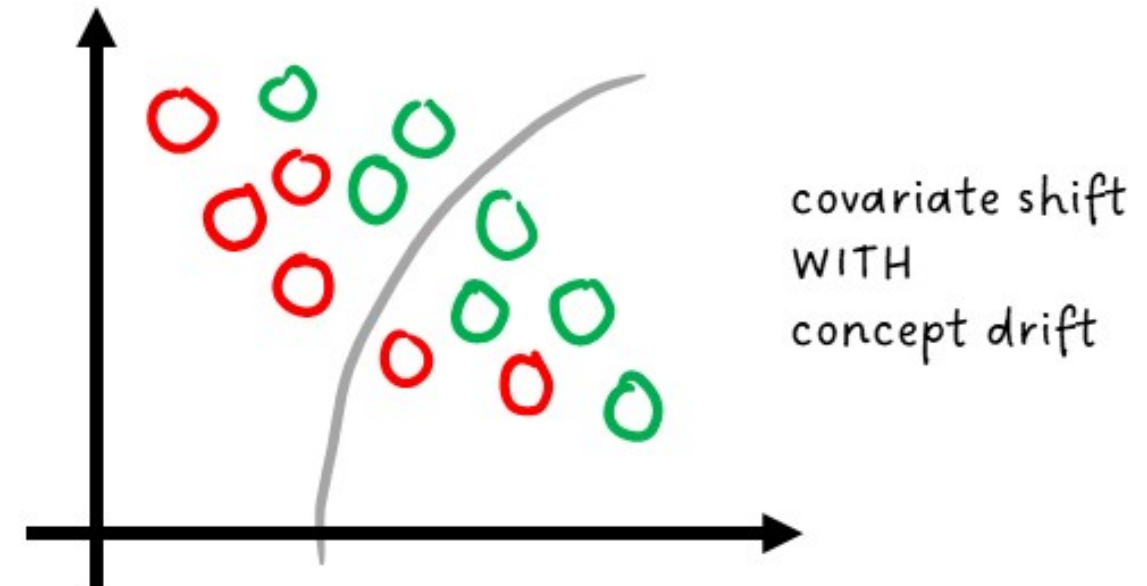
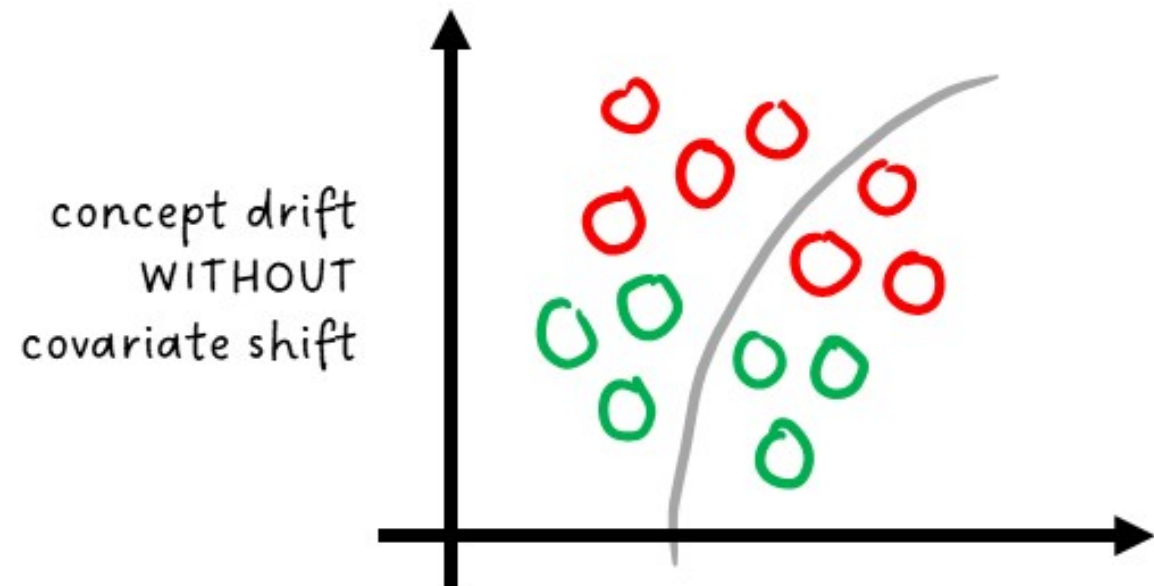
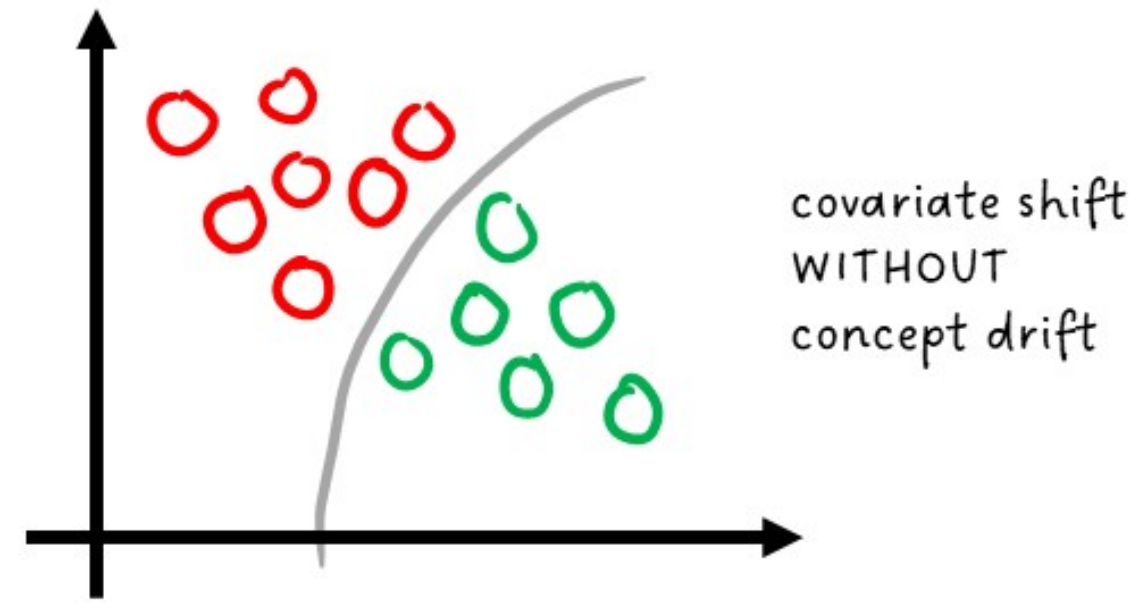
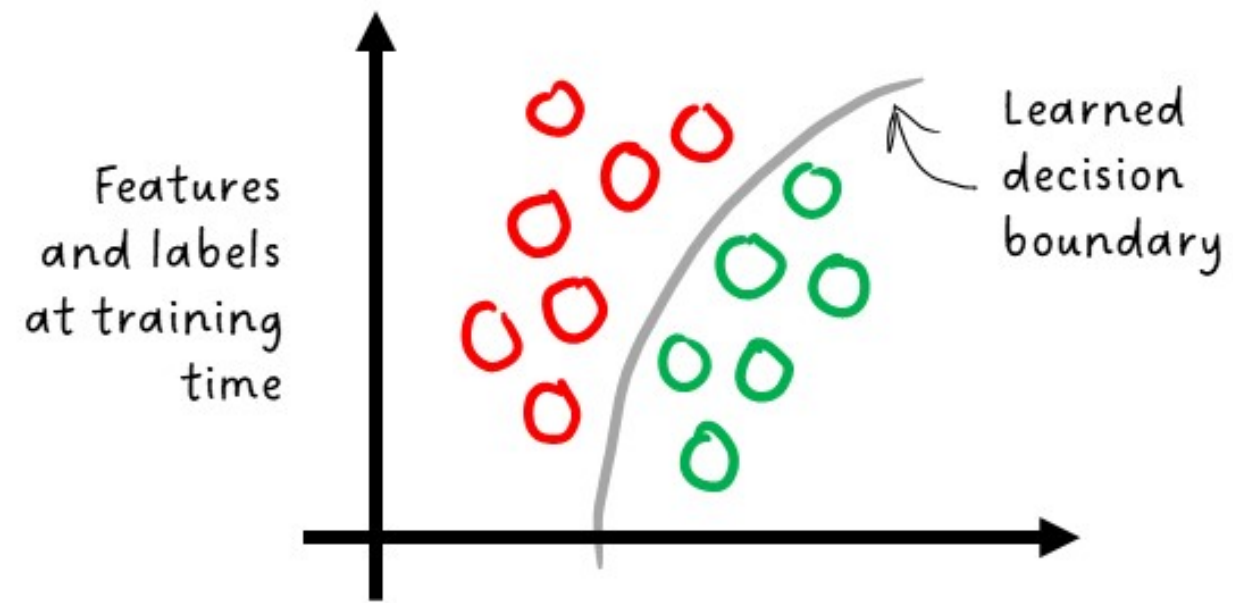
Indirect approach:

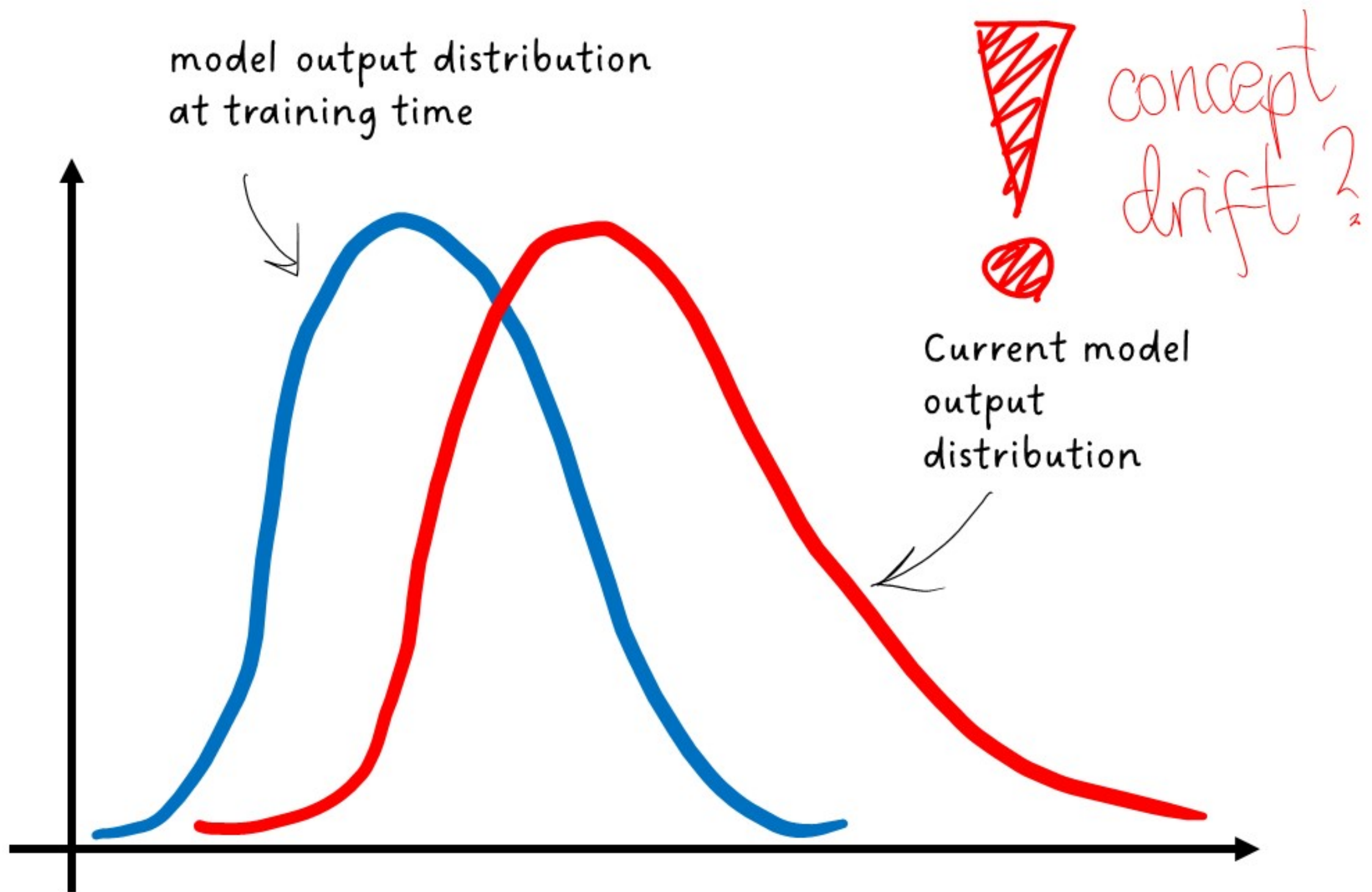
Monitor what you have: input features!



* features == "covariates" in statistical jargon

Limitations of input monitoring





Let's practice!

MLOPS DEPLOYMENT AND LIFE CYCLING

Monitoring and alerting

MLOPS DEPLOYMENT AND LIFE CYCLING



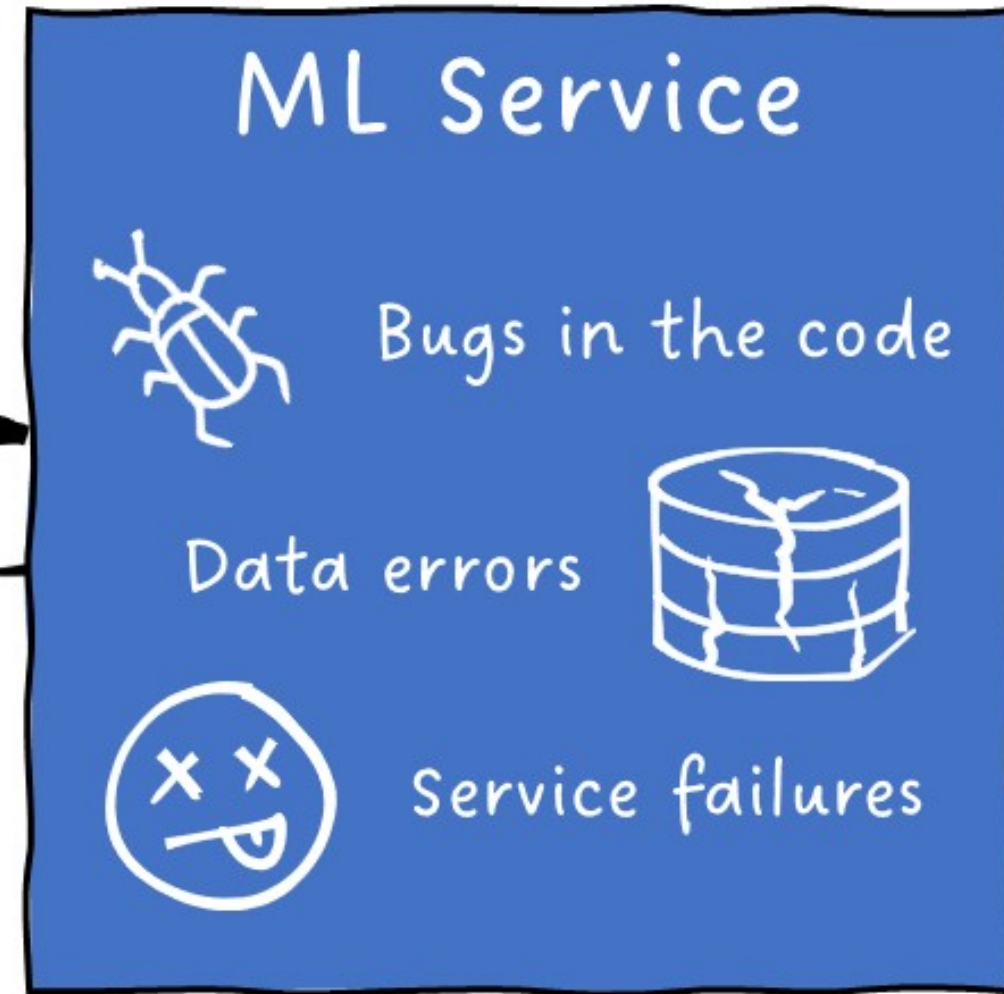
Nemanja Radojkovic

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OUTSIDE
WORLD

ML Service

OUTSIDE
WORLD



OUTSIDE
WORLD

*monitoring &
alerting system

ML Service

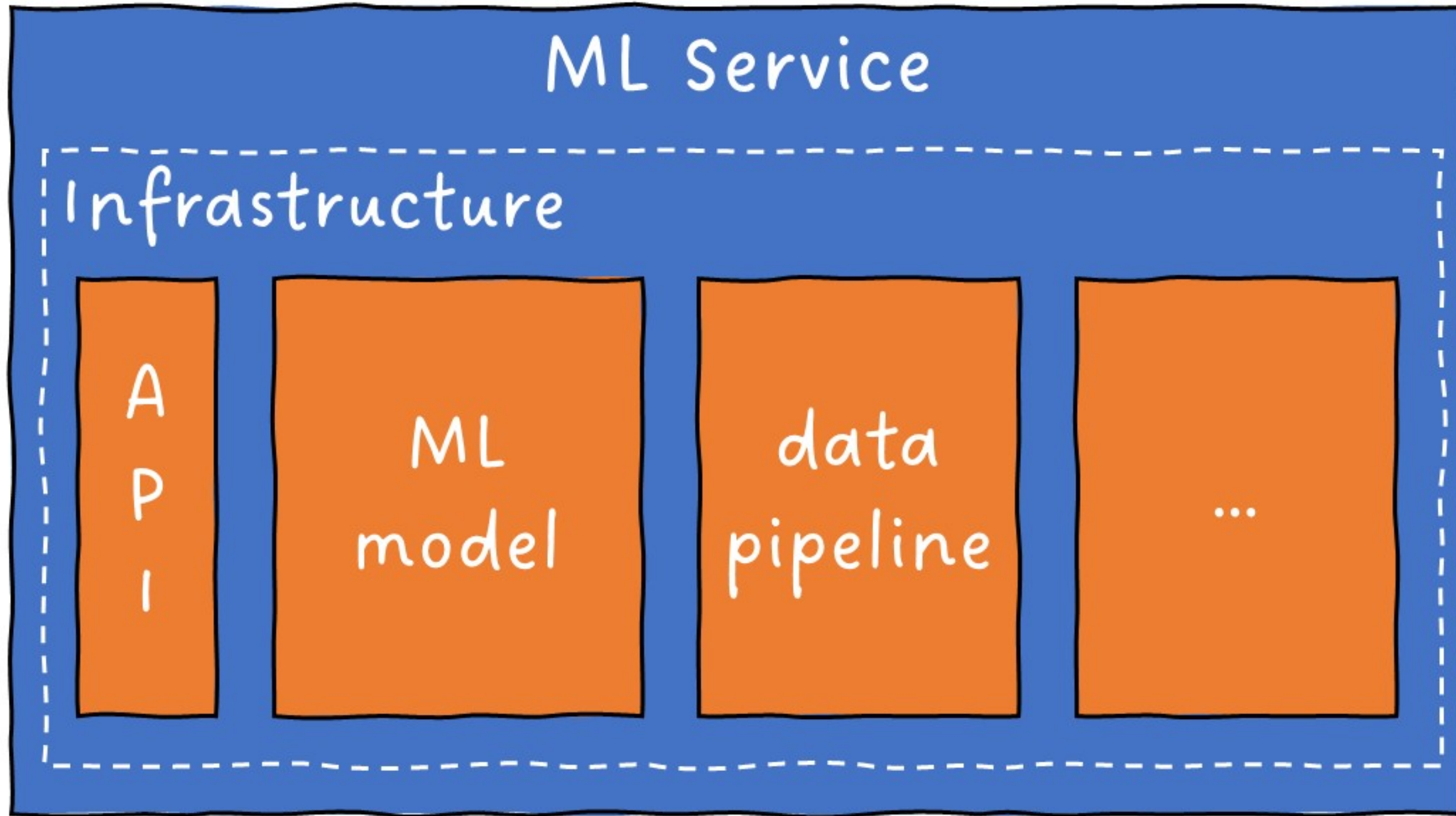


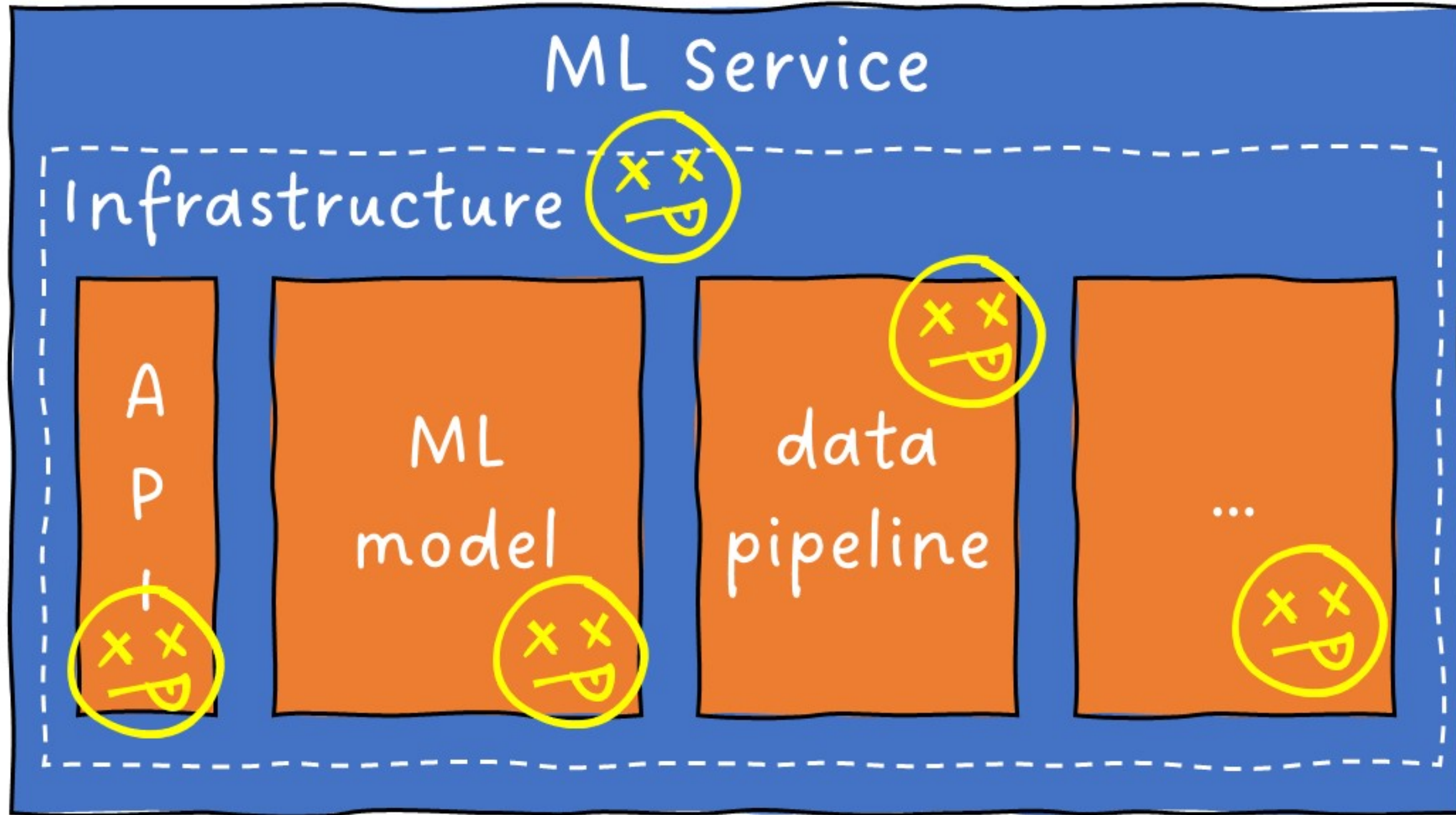
Bugs in the code

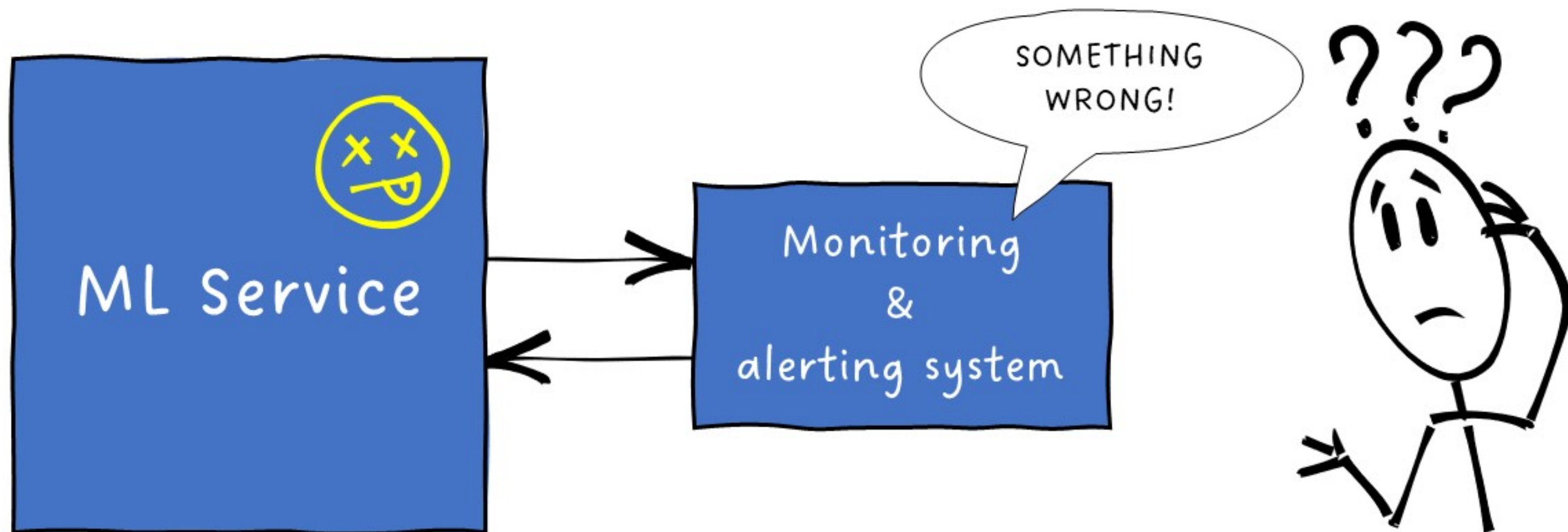
Data errors

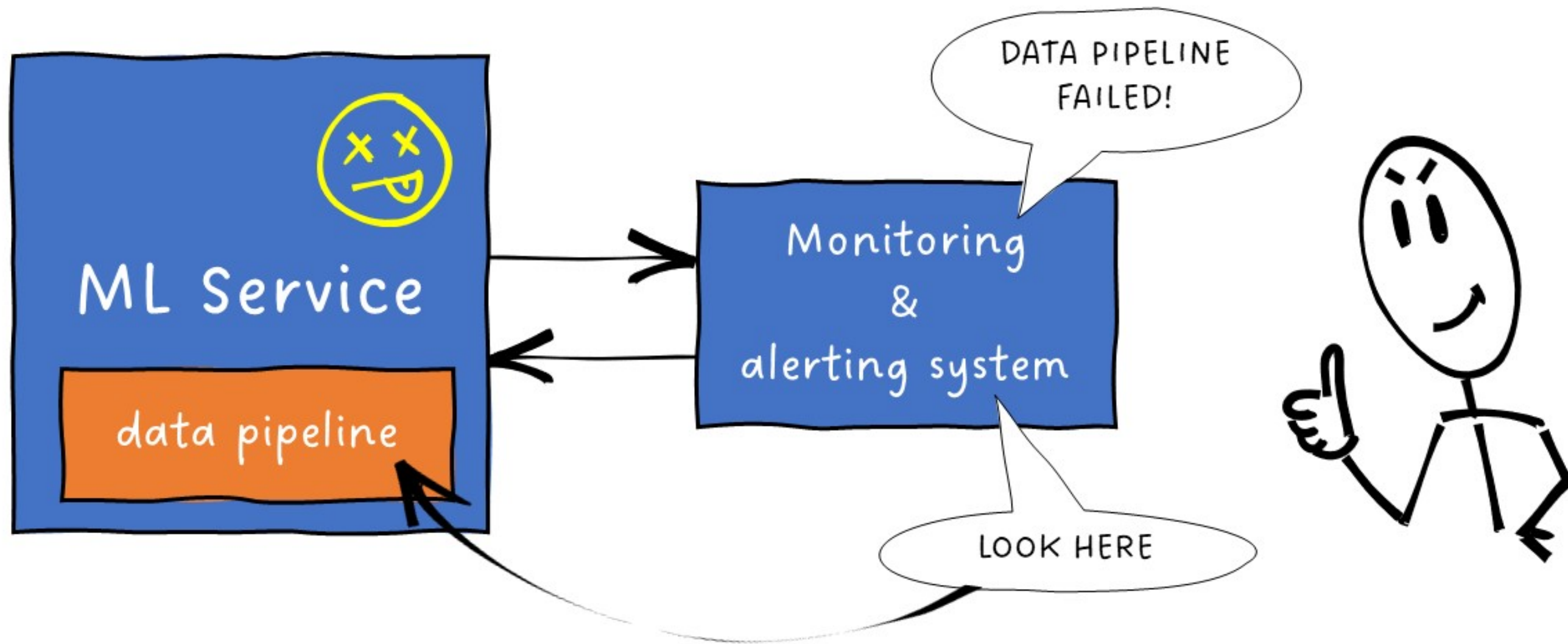


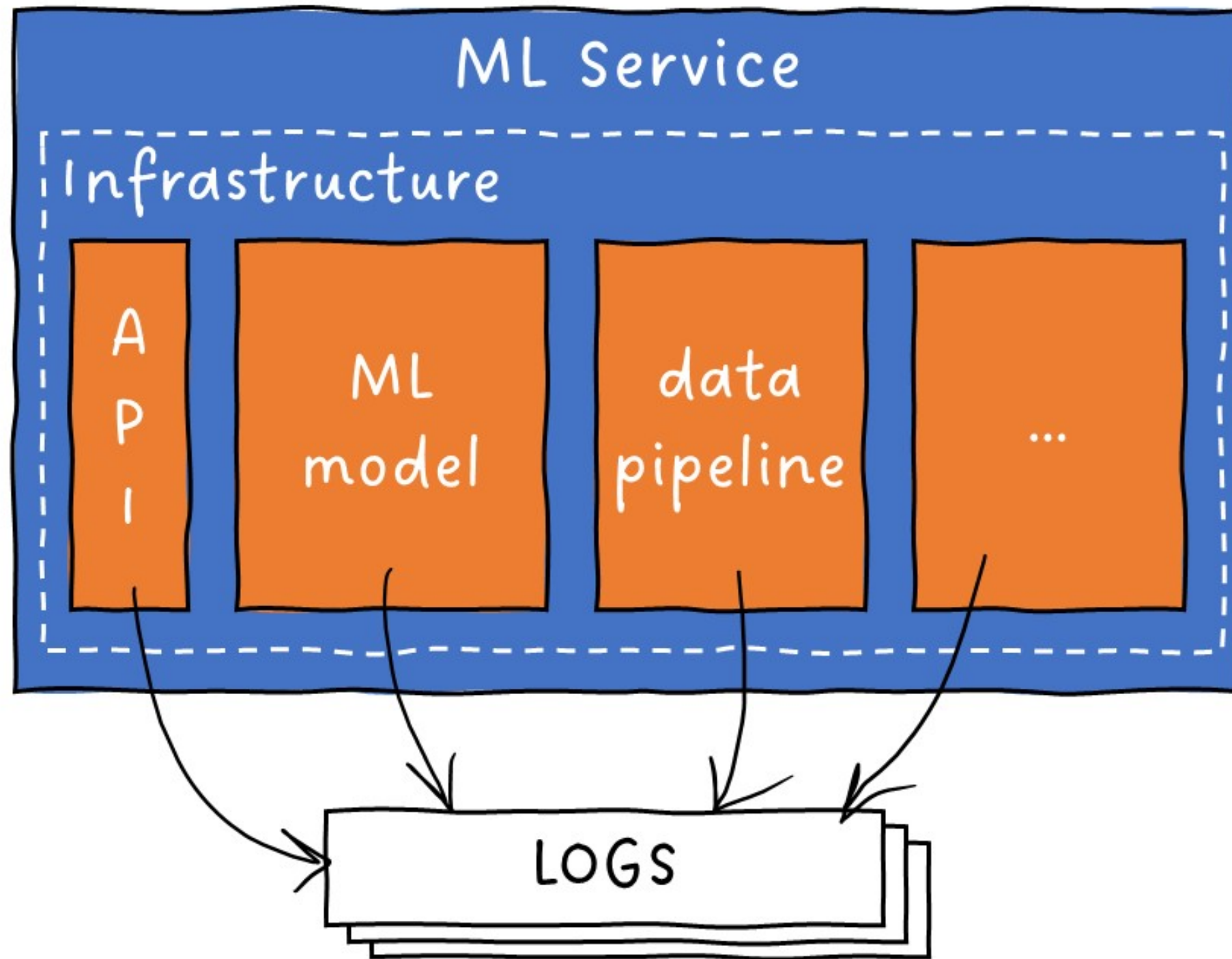
Service failures

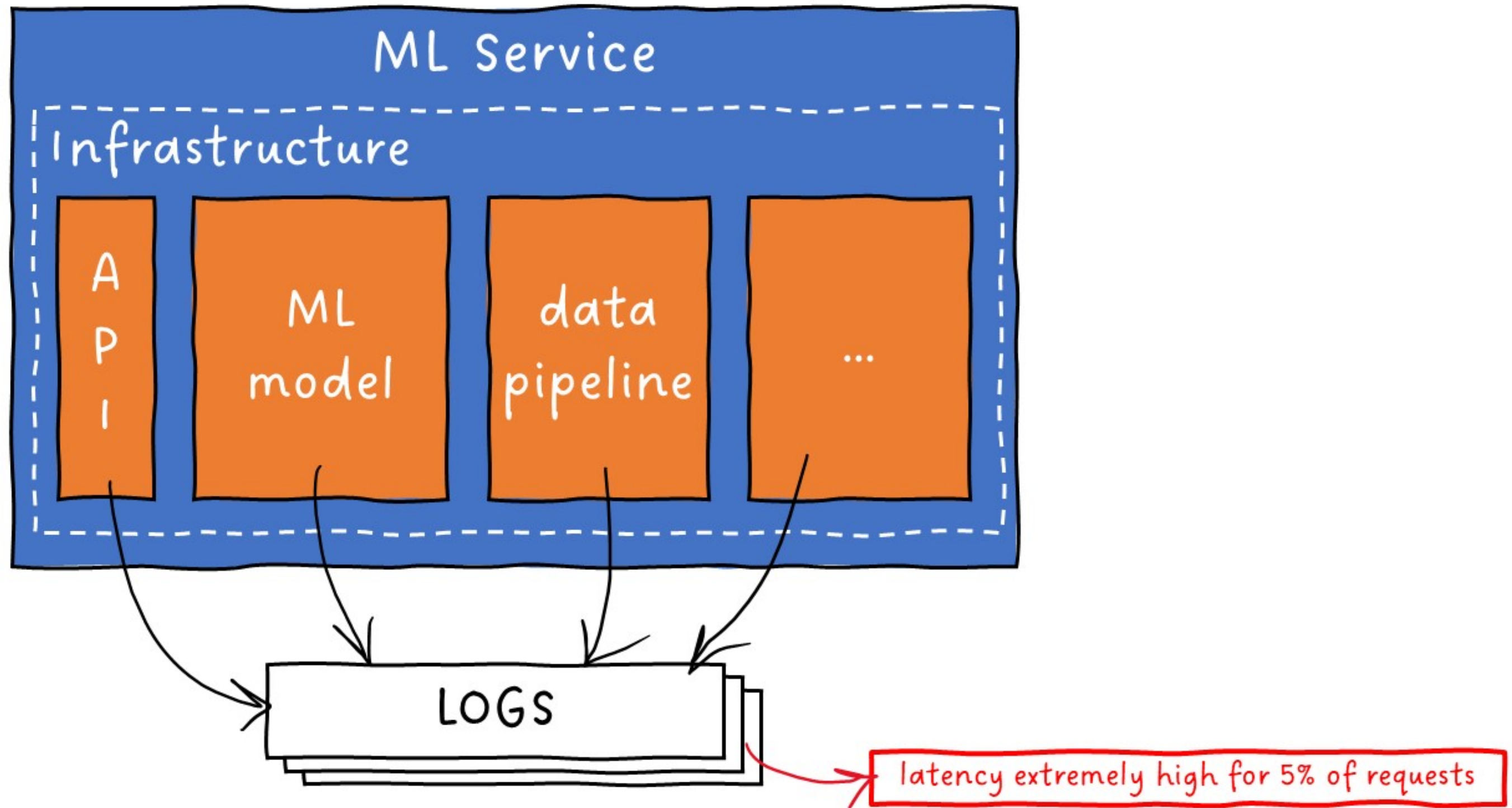


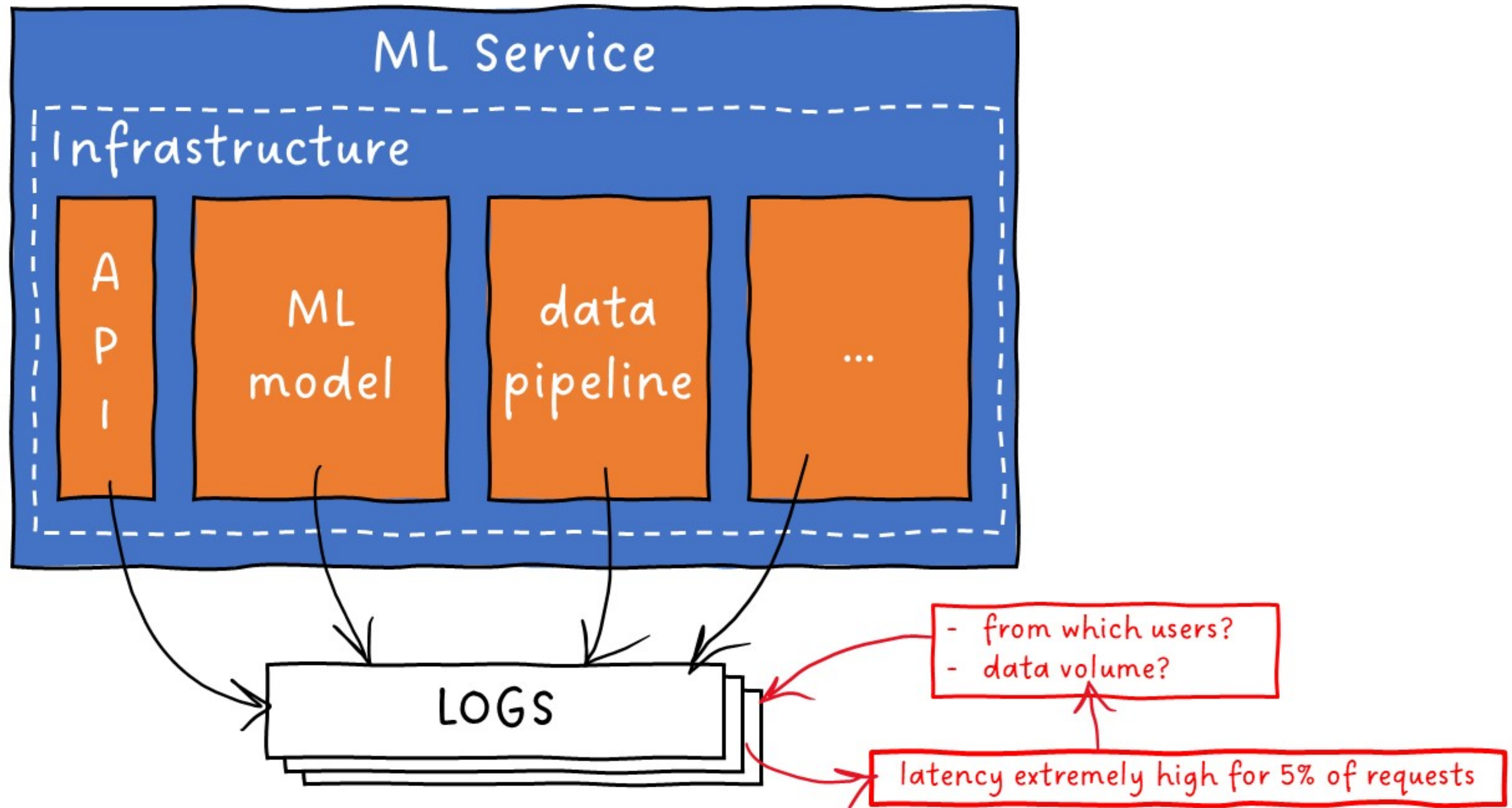




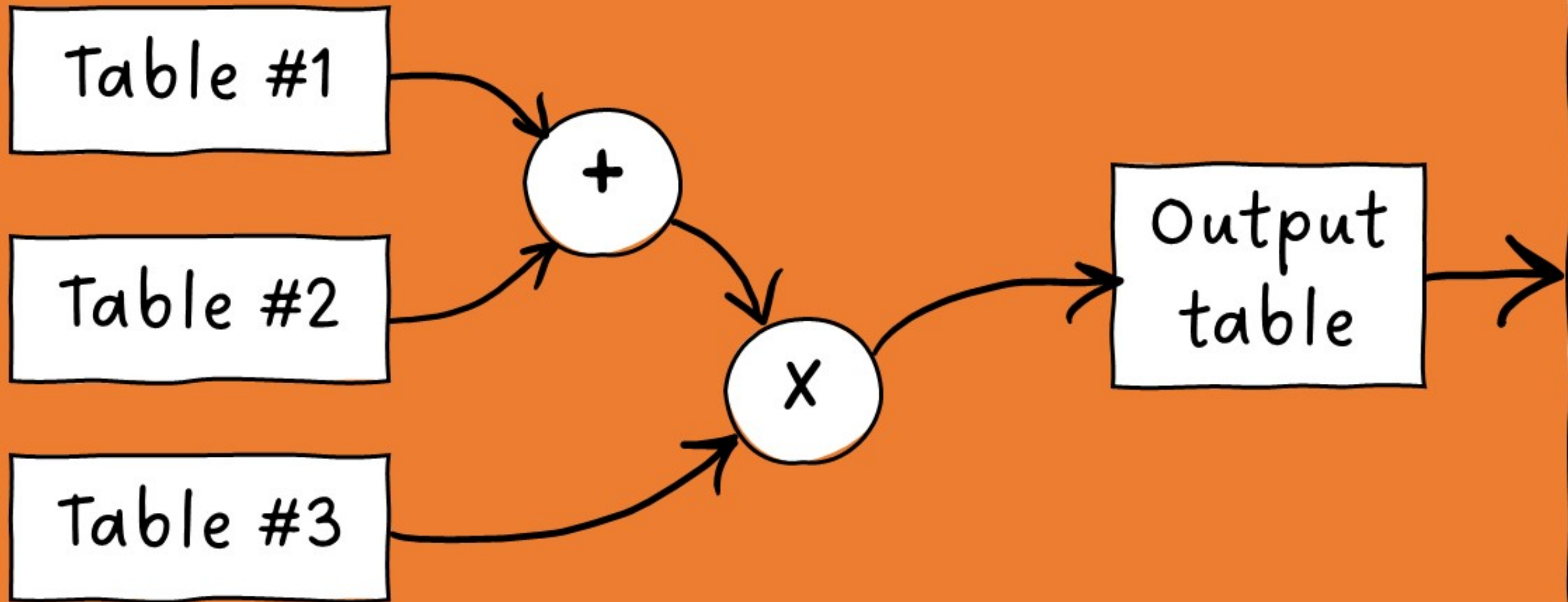




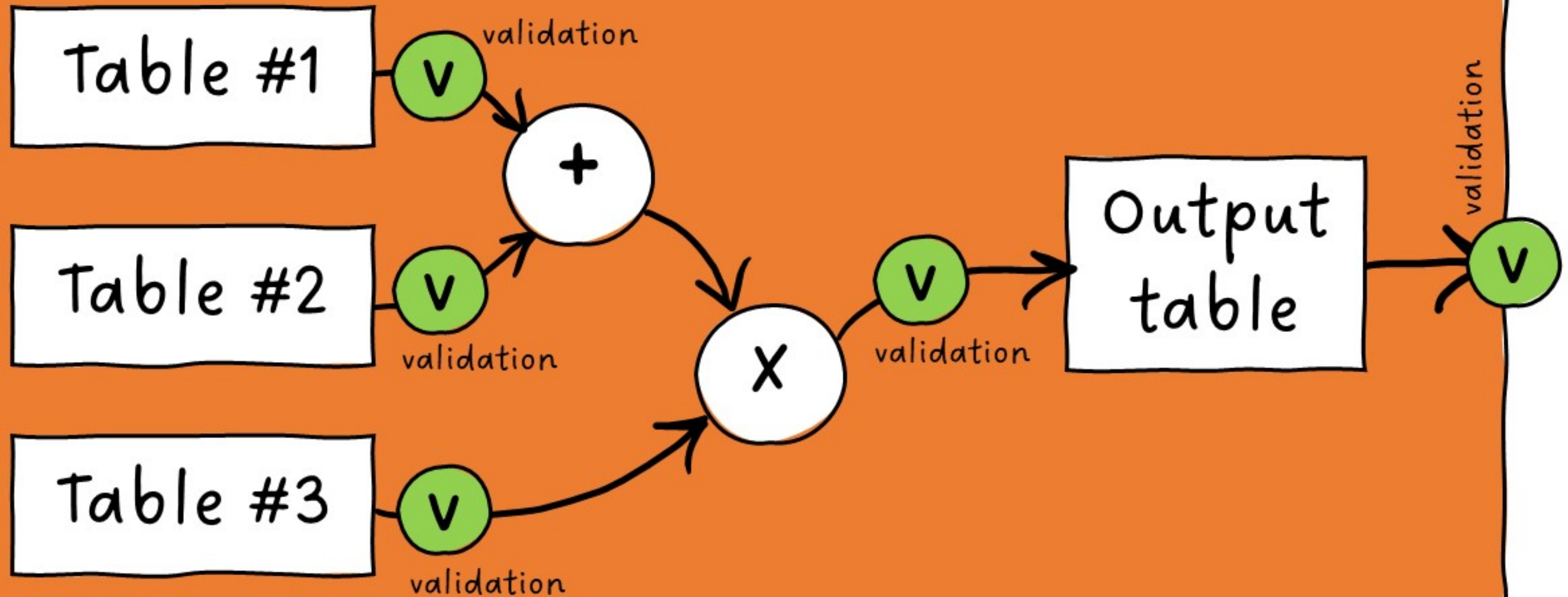




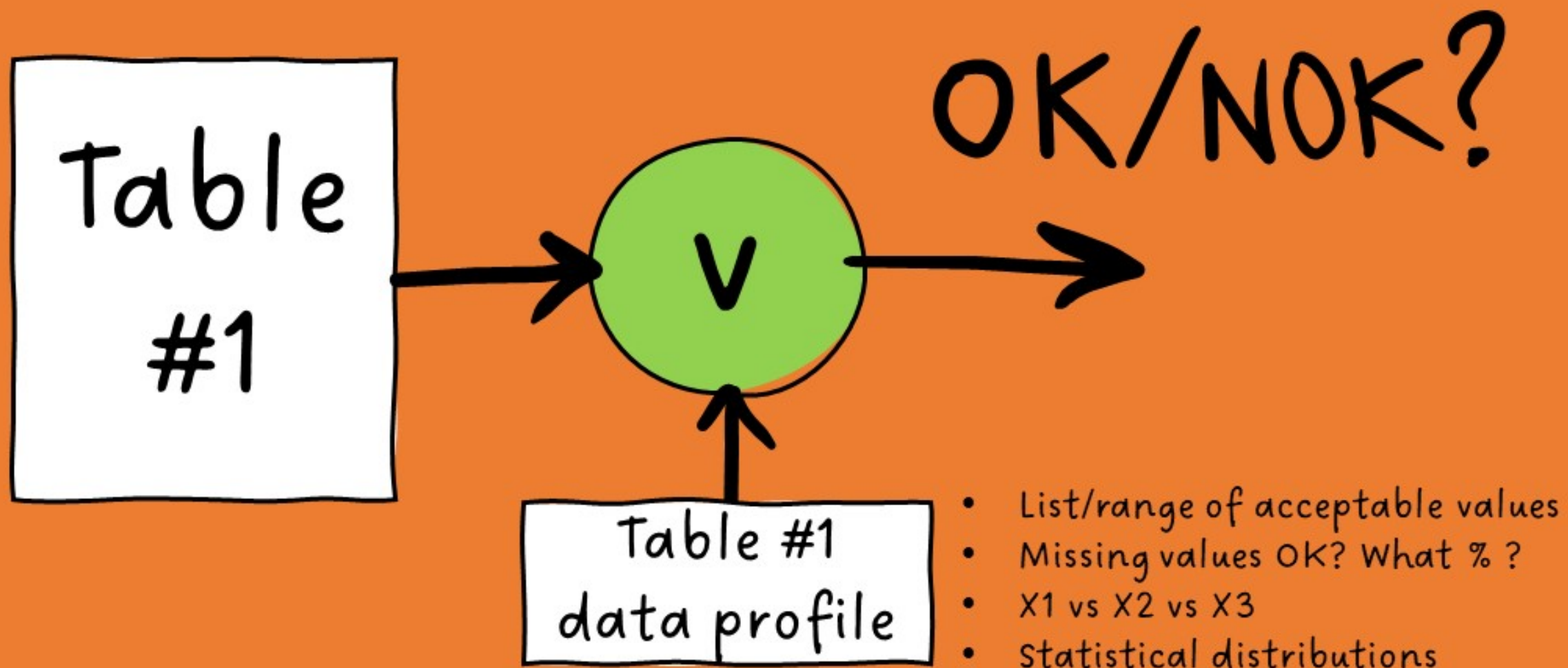
data pipeline



data pipeline



data validation



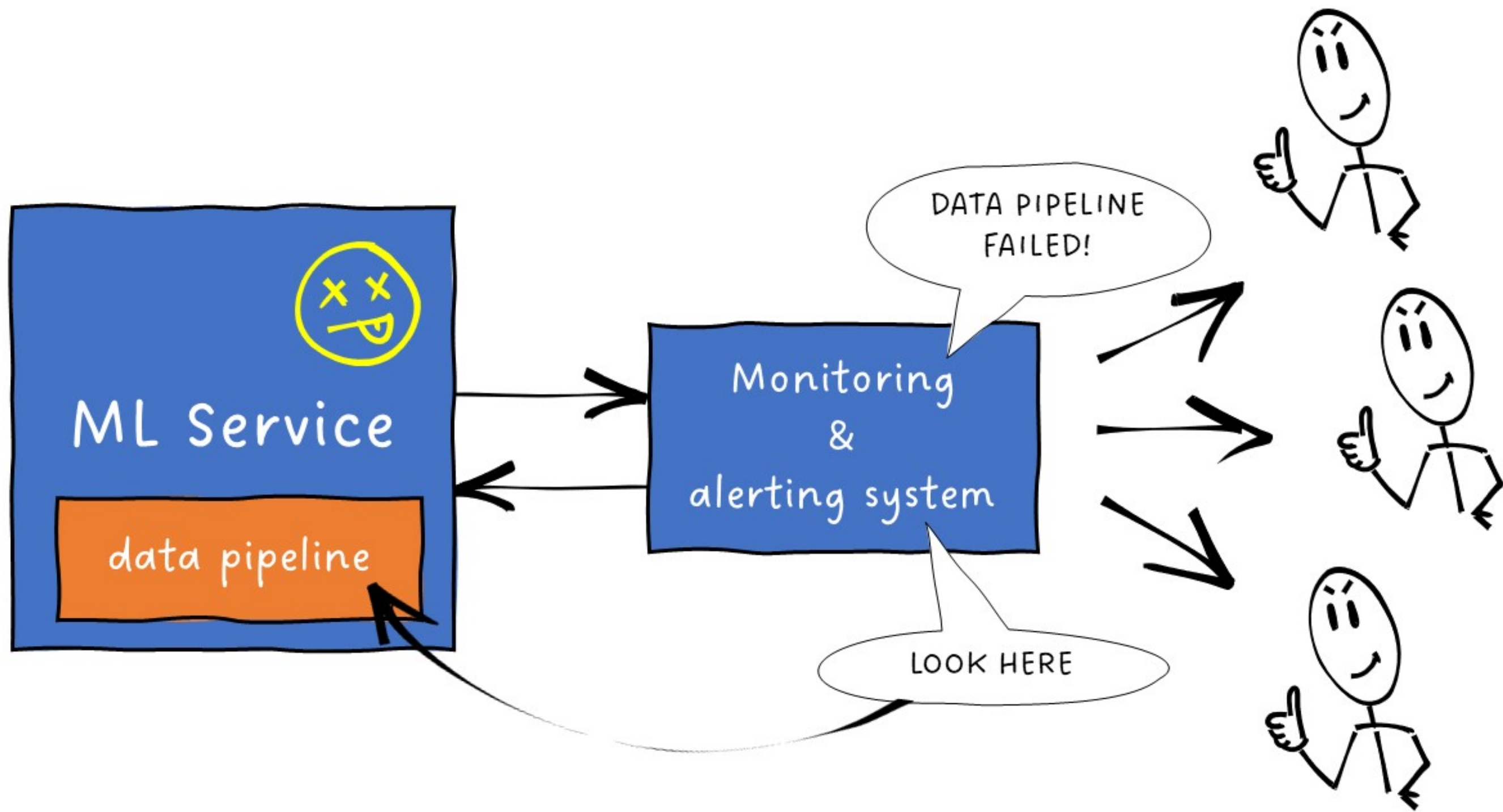
Statistical validation

Can be:

- too sensitive
- not informative enough

Risk

- Too many alerts
- "Alert fatigue"
- Important alerts going unnoticed



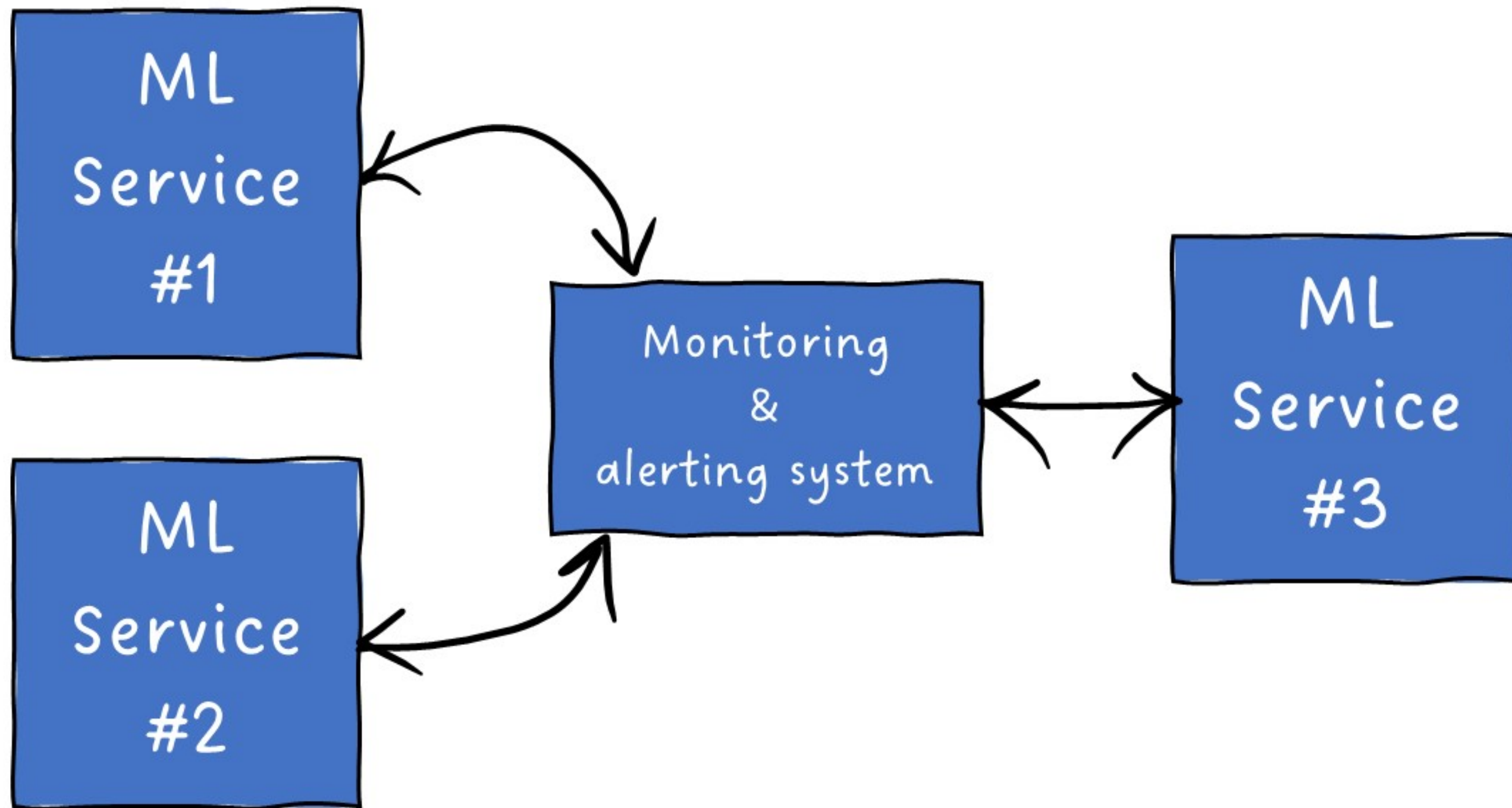
Learn from your history

After treating the incident => Record root cause and resolution steps

Example from Google^[1]:

- 10 years of incidents recorded and analyzed
- > 2/3 were not ML-related!

¹ How ML Breaks: A Decade of Outages for One Large ML Pipeline,
<https://www.usenix.org/conference/opml20/presentation/papasian>



Let's practice!

MLOPS DEPLOYMENT AND LIFE CYCLING

Model maintenance

MLOPS DEPLOYMENT AND LIFE CYCLING



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Previously...

- How to catch anomalies and disturbances?

Focus of this lesson: ML model failure

Model
performance
deteriorated?

Model
performance
deteriorated?



Improve the
MODEL

Model
performance
deteriorated?



Improve the
MODEL



Improve the
DATASET

Model
performance
deteriorated?



Improve the
MODEL



Improve the
DATASET

MODEL-centric approach to
ML development

- typical for competitions, where dataset is fixed
- try new models
- try new DERIVED features

Model
performance
deteriorated?



Improve the
MODEL

MODEL-centric approach to
ML development

- typical for competitions, where dataset is fixed
- try new models
- try new DERIVED features



Improve the
DATASET

In real life: Freedom to
clean, enrich, enlarge
dataset.

DATA-centric approach to
ML development

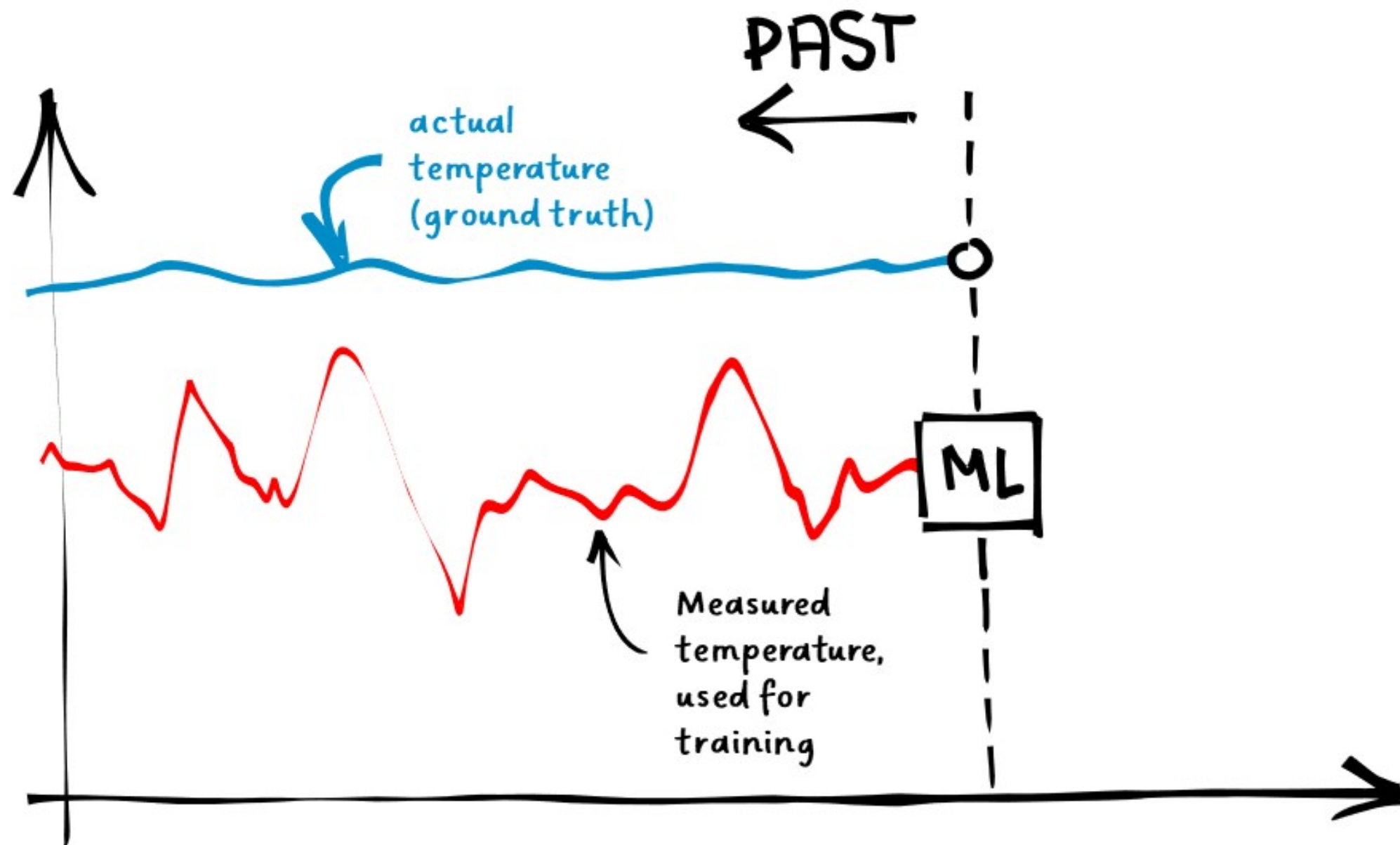
¹ <https://datacentricai.org/>

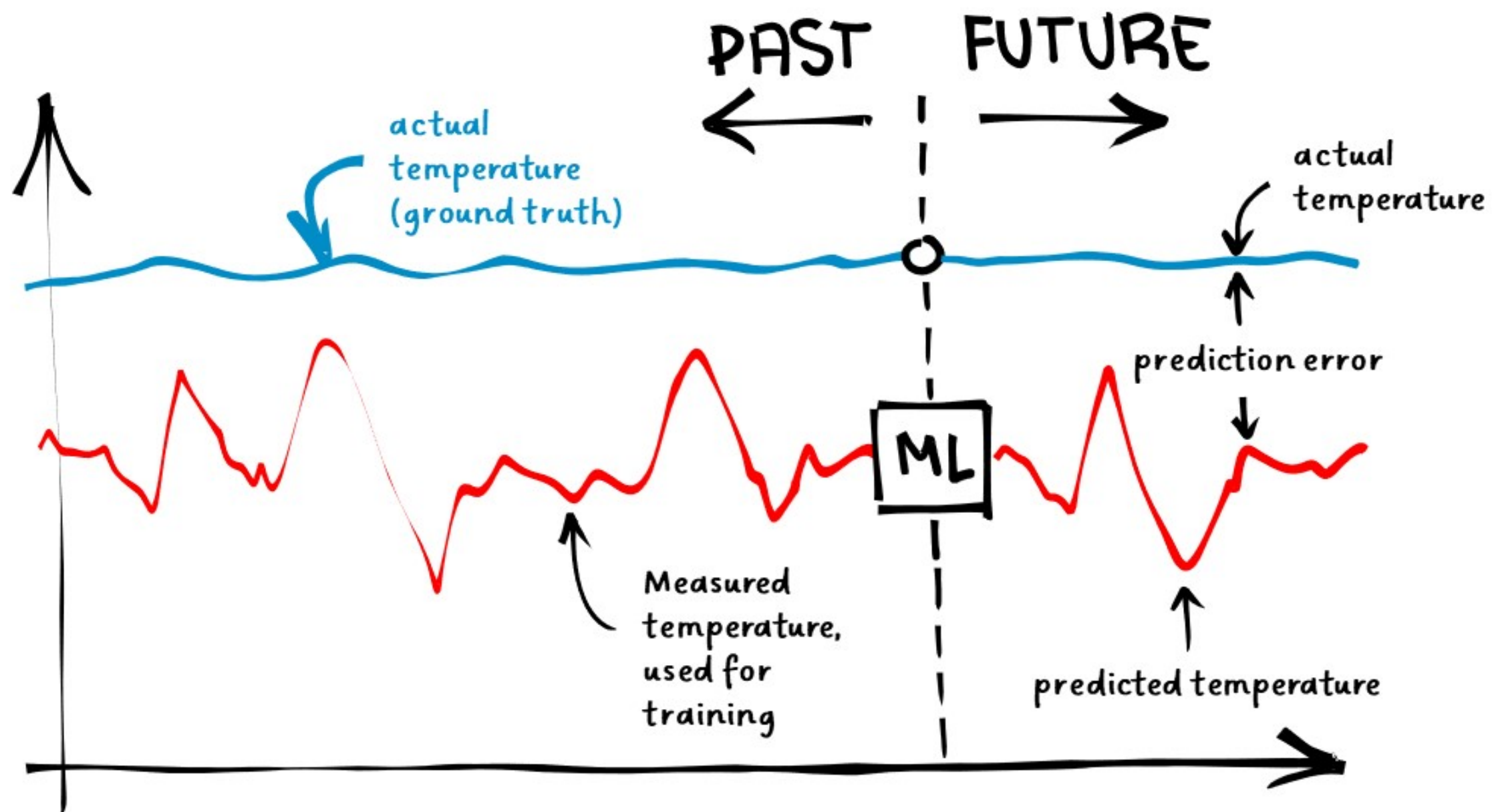
Quality above quantity

- Get more features with relevant information
- Get better labels

Label == Target variable value in the training set

Label quality == label closeness to the ground truth



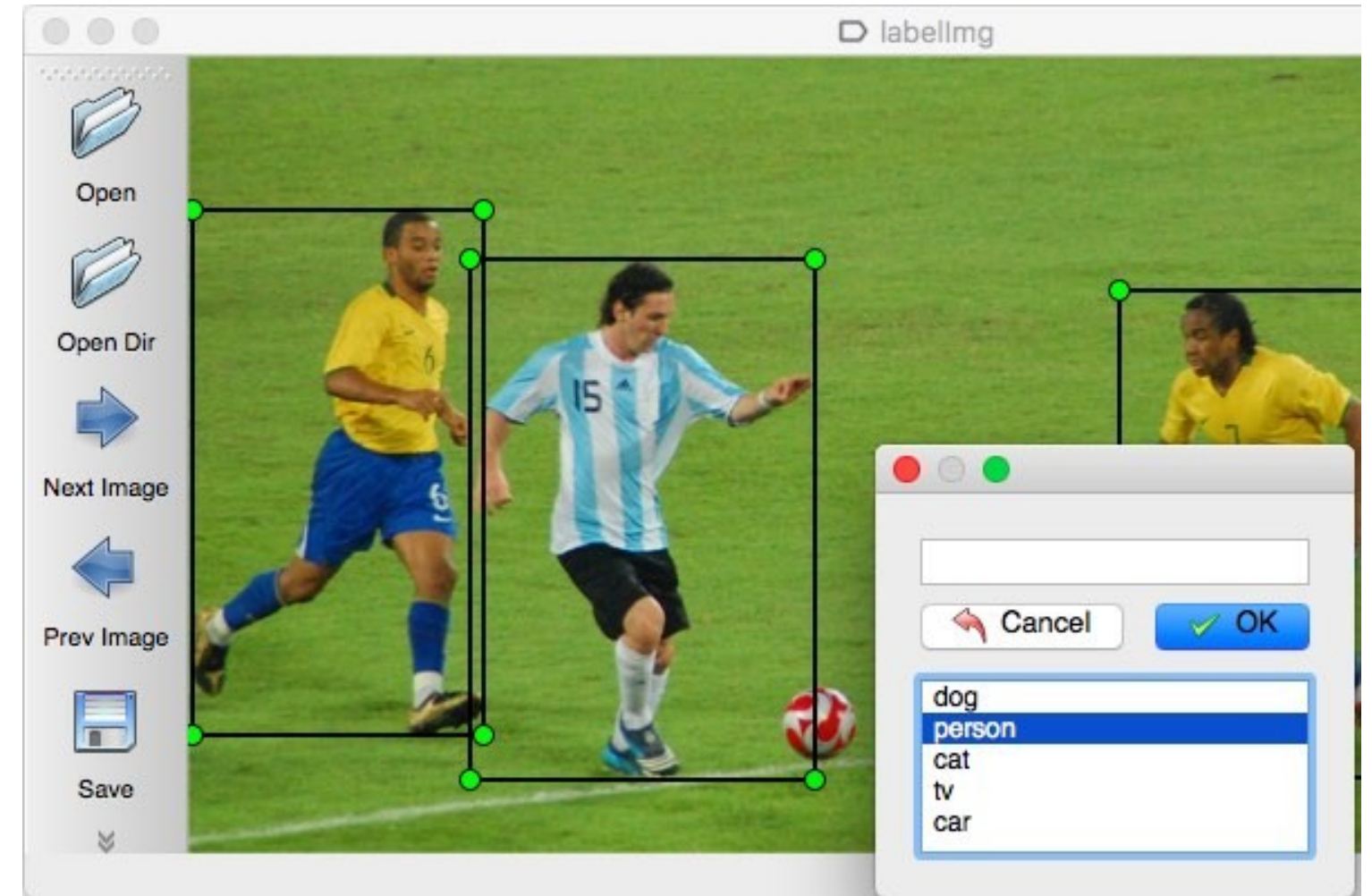


Benefits of labeling tools

Manual labeling is complex, lengthy, error-prone

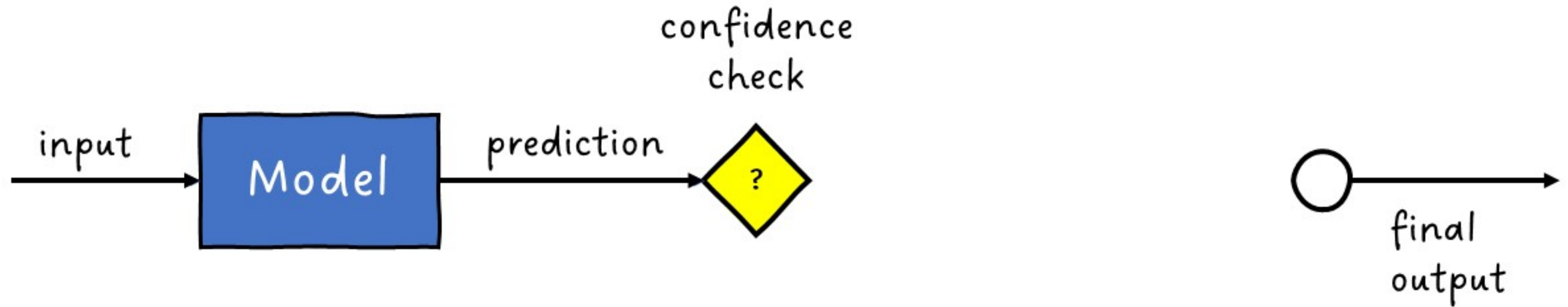
Use good labeling tools!

- Labeling more efficient
- More accurate
- User interface fit for purpose
- "Label these examples first for maximum impact"
- "It seems you made a mistake here"

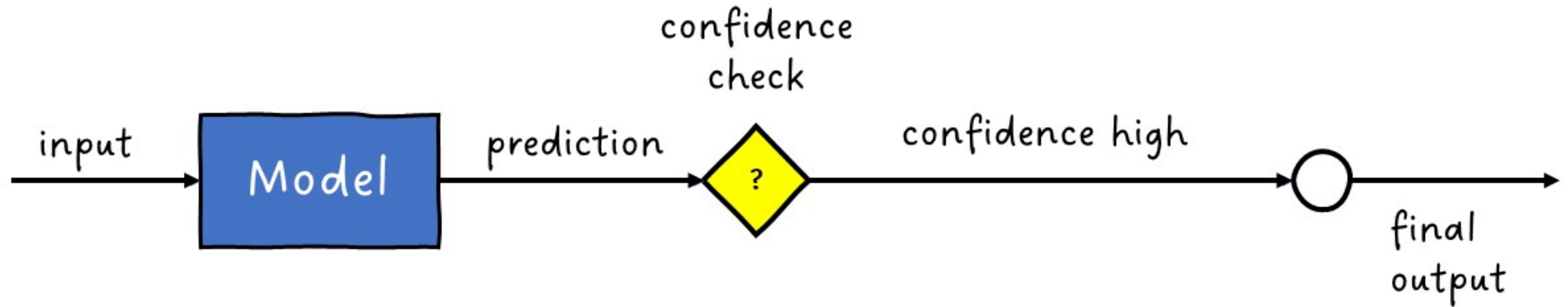


Example: Image labeling tool, for building image classifiers

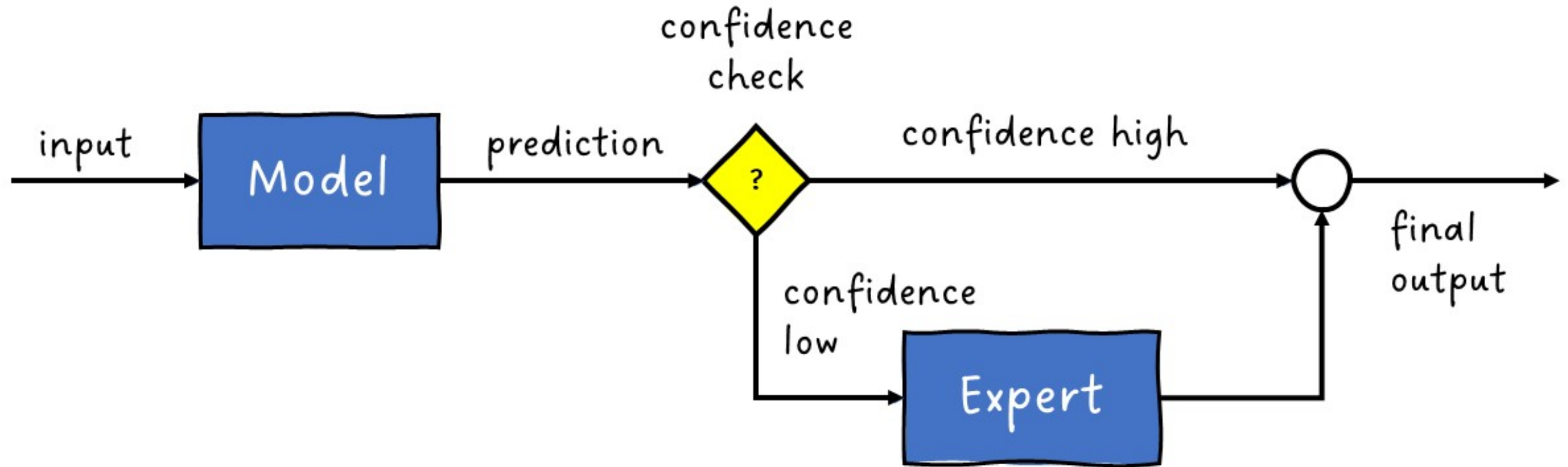
Human-in-the-loop system



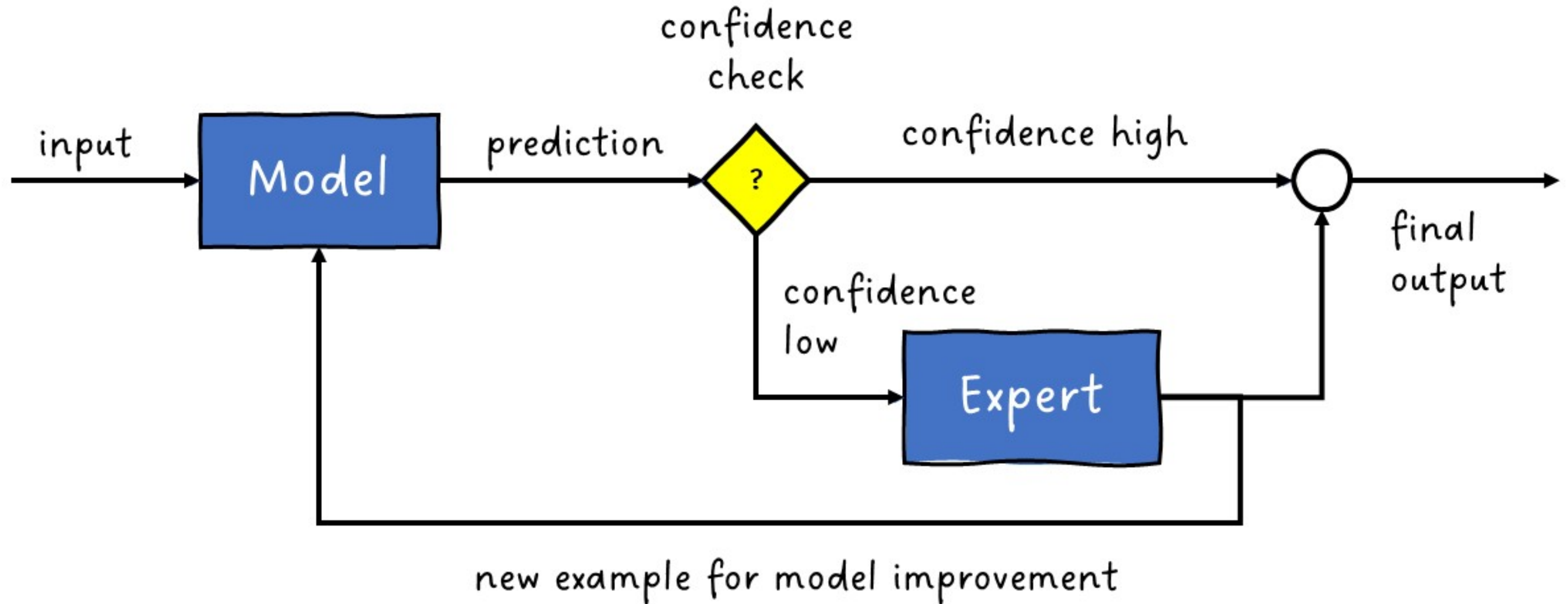
Human-in-the-loop system



Human-in-the-loop system



Human-in-the-loop system



When the labels arrive

- Run ML build pipeline => New model
- Check if new model is better than the old one
 - YES => test and deploy
 - NO => keep searching,

Keep searching

- Try new models
- Try new features
- Try new data sources

Immensely helpful: Metadata store (MLFlow Tracking, etc)

- Document the model selection journey
- Avoid repeating same experiments

In any case: *MLOps helps us maintain our model in the fastest, most efficient way*

Let's practice!

MLOPS DEPLOYMENT AND LIFE CYCLING

Model governance

MLOPS DEPLOYMENT AND LIFE CYCLING



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ArtiQ Wins UK Government Funding for its AI Algorithms to Support Timely Diagnosis of Lung Diseases

by ArtiQ Jun 16, 2021

Business Of Law ; Internet Law & Cyber-Security

Law Bots: How AI Is Reshaping the Legal Profession



10 Min Read

By: Matthew Stepka | February 21, 2022

Artificial Intelligence and Machine Learning

9 min read

5 Ways AI is Transforming the Finance Industry

Check out the easy ways by which artificial intelligence can transform the finance industry.



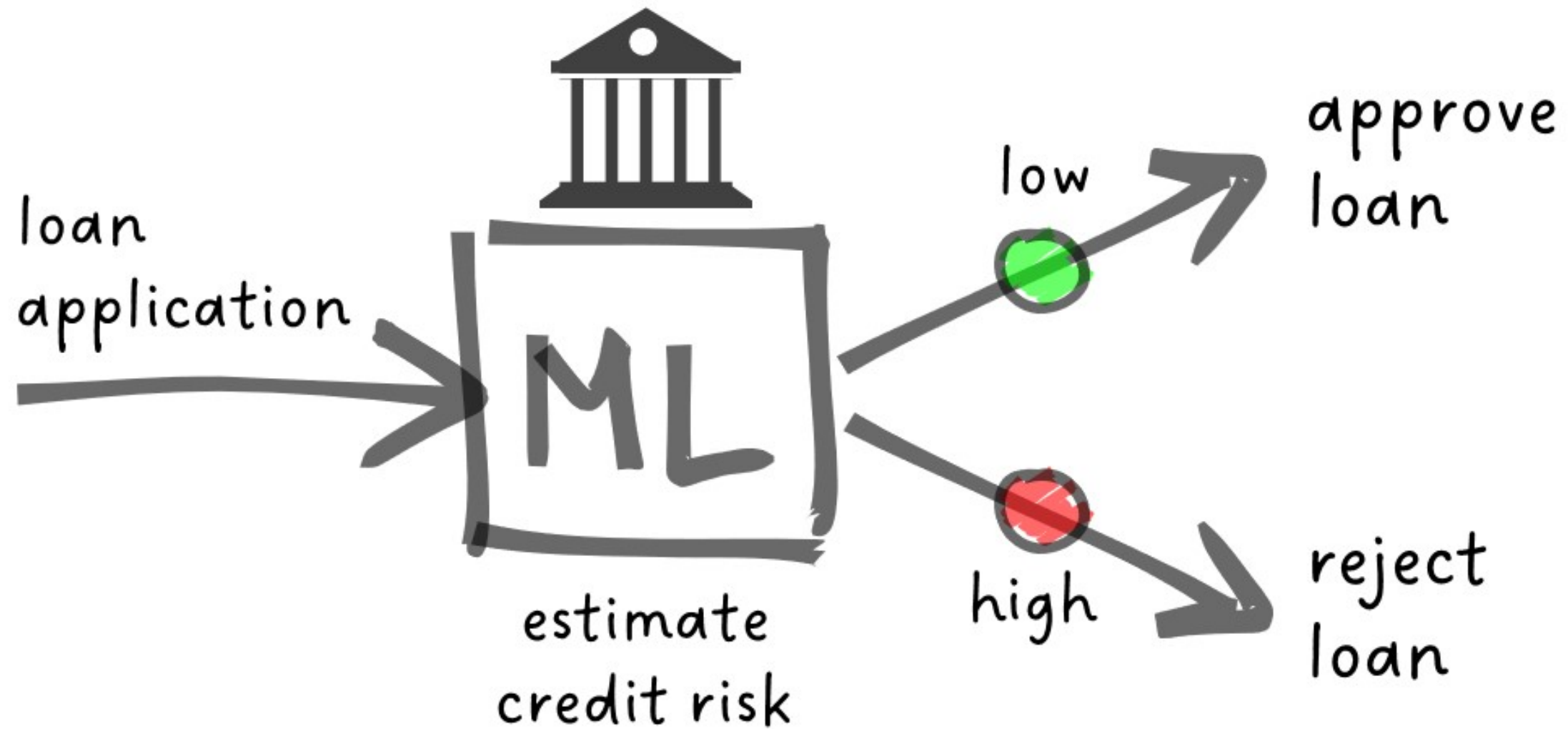
Pinakin Ariwala

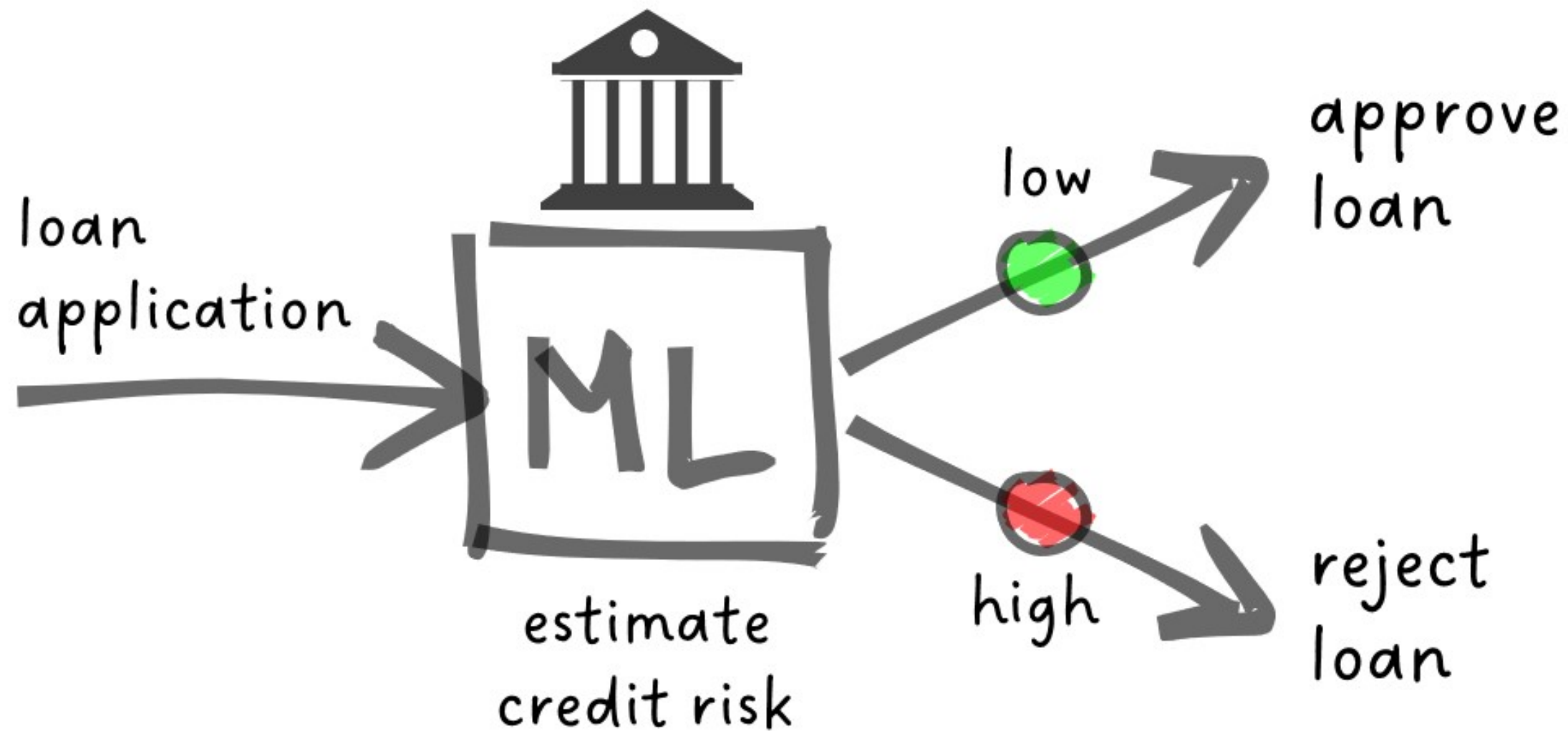
Updated on Oct 14

2 minute read · November 30, 2022 5:29 PM GMT+1 · Last Updated a day ago

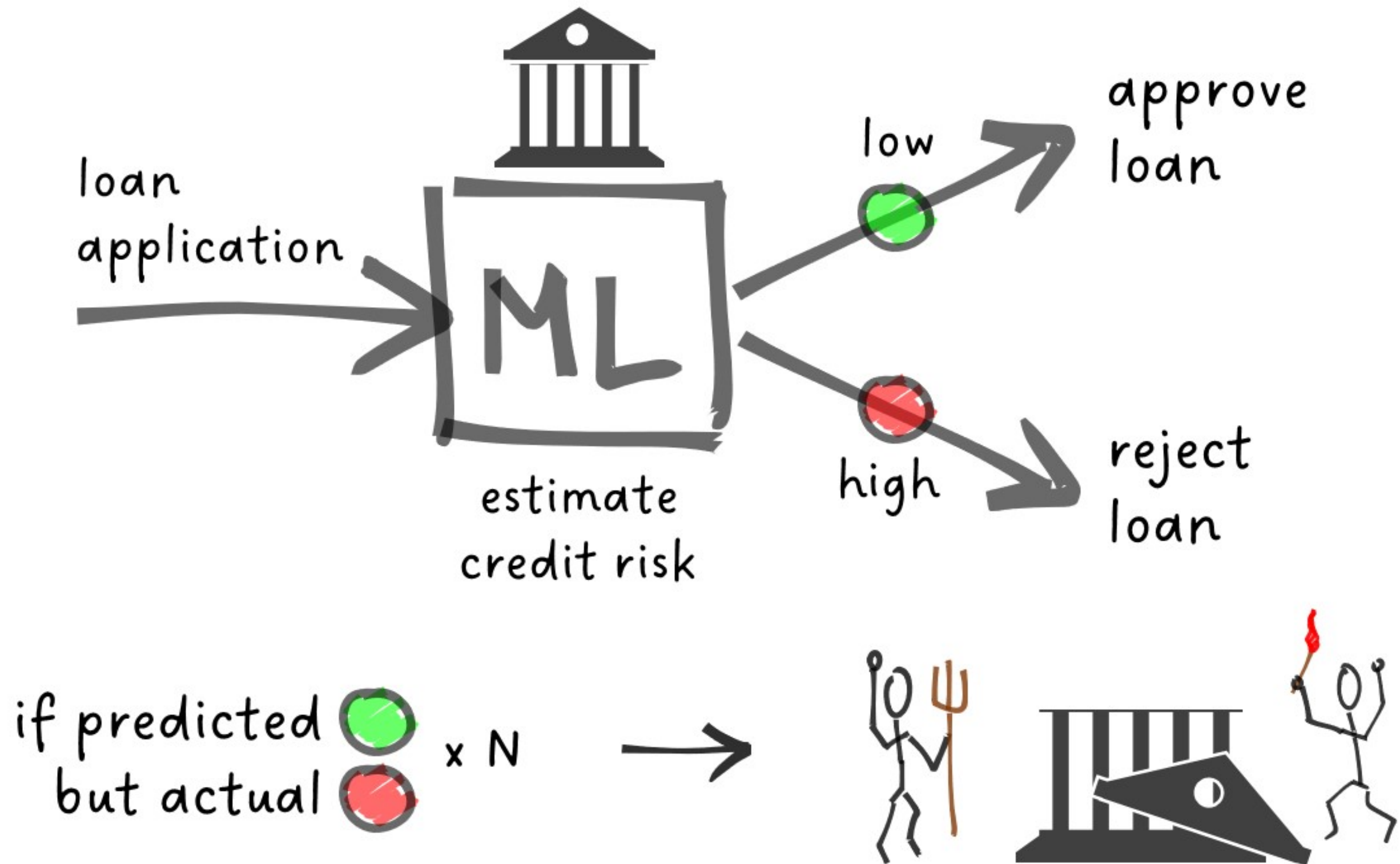
Honda to develop advanced level 3 self-driving technology by 2029

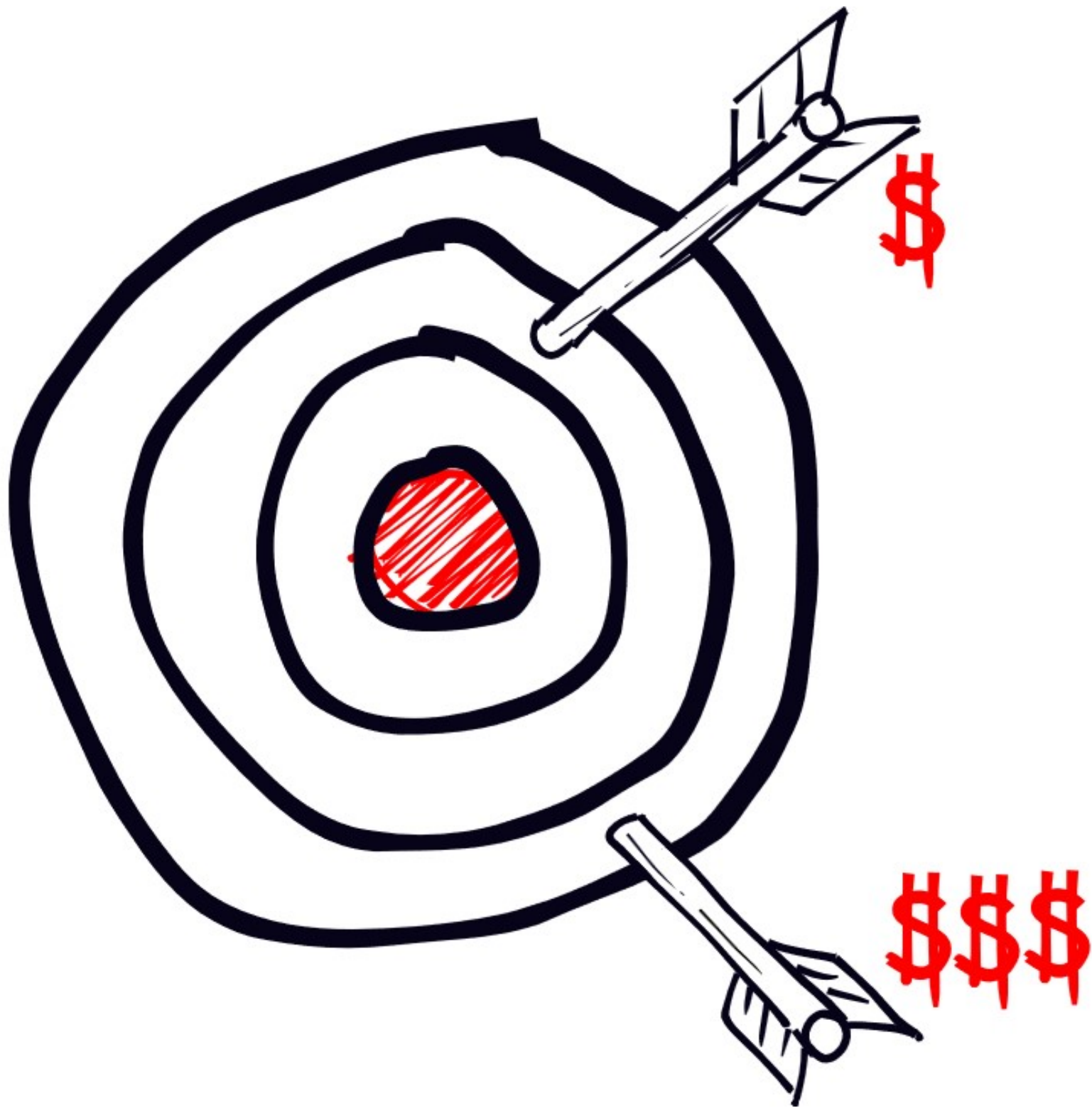
Reuters

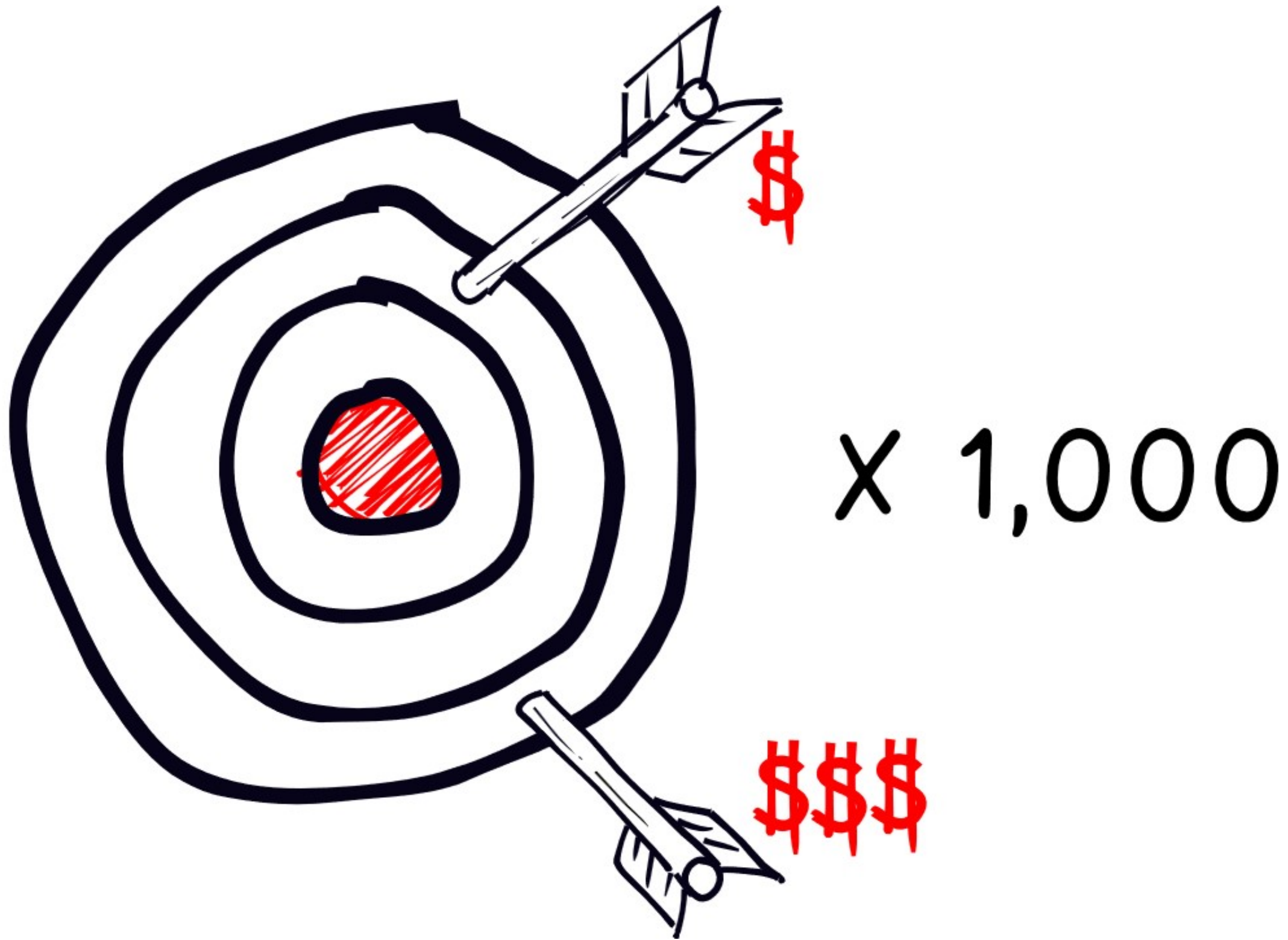




if predicted  x N
but actual 





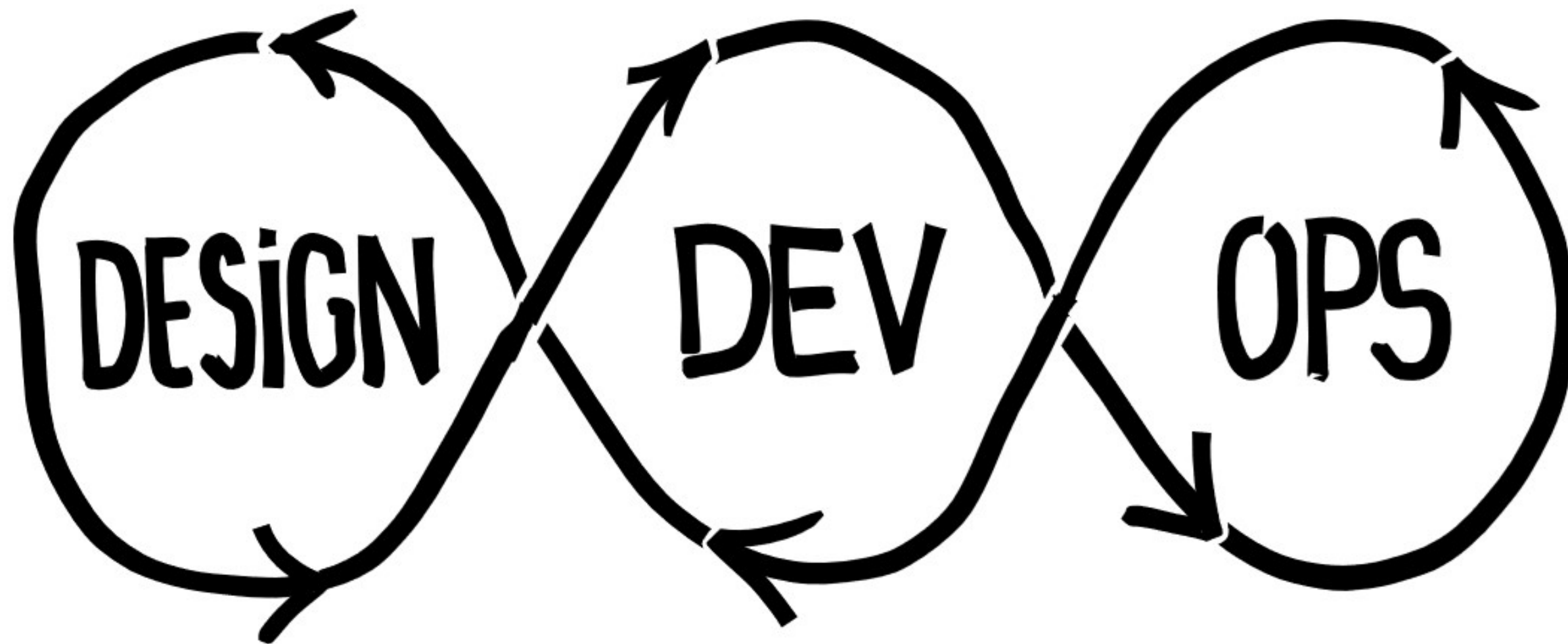


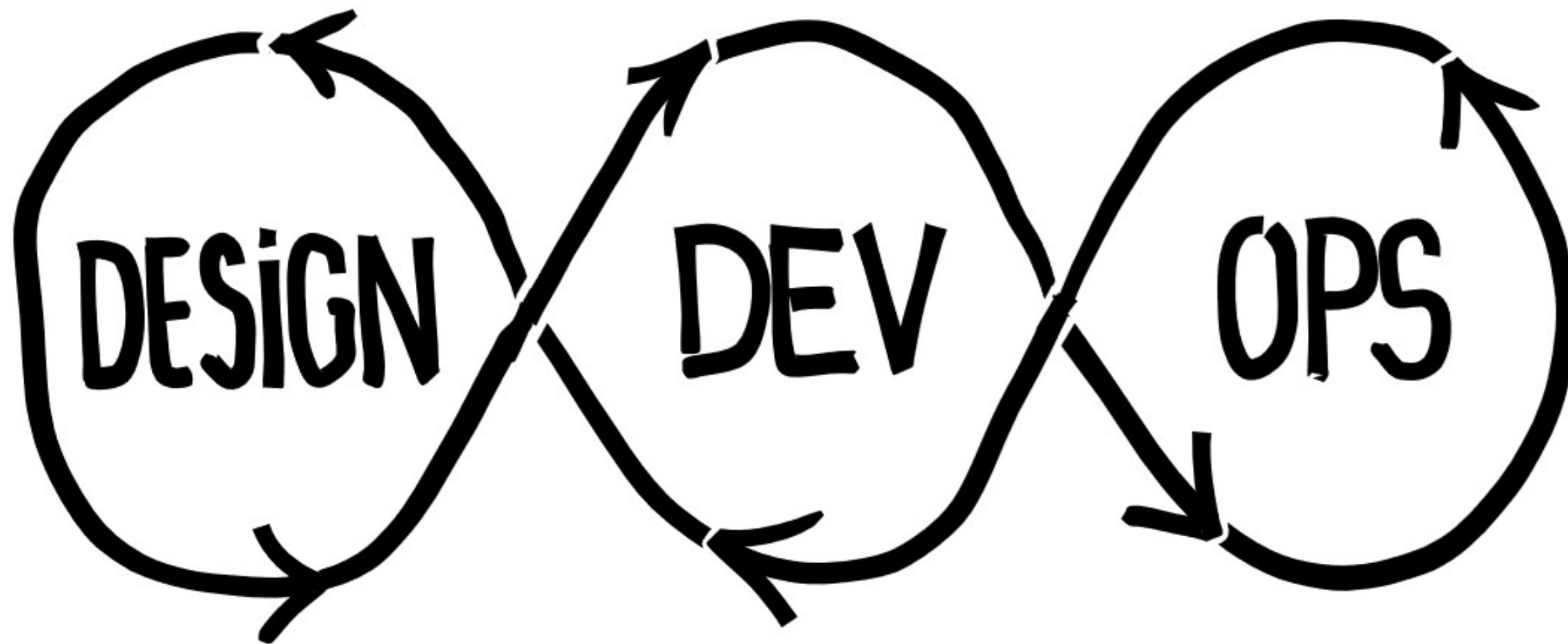
Governance

AI/ML model governance is the overall process for how an organization controls access, implements policy, and tracks activity for models and their results.

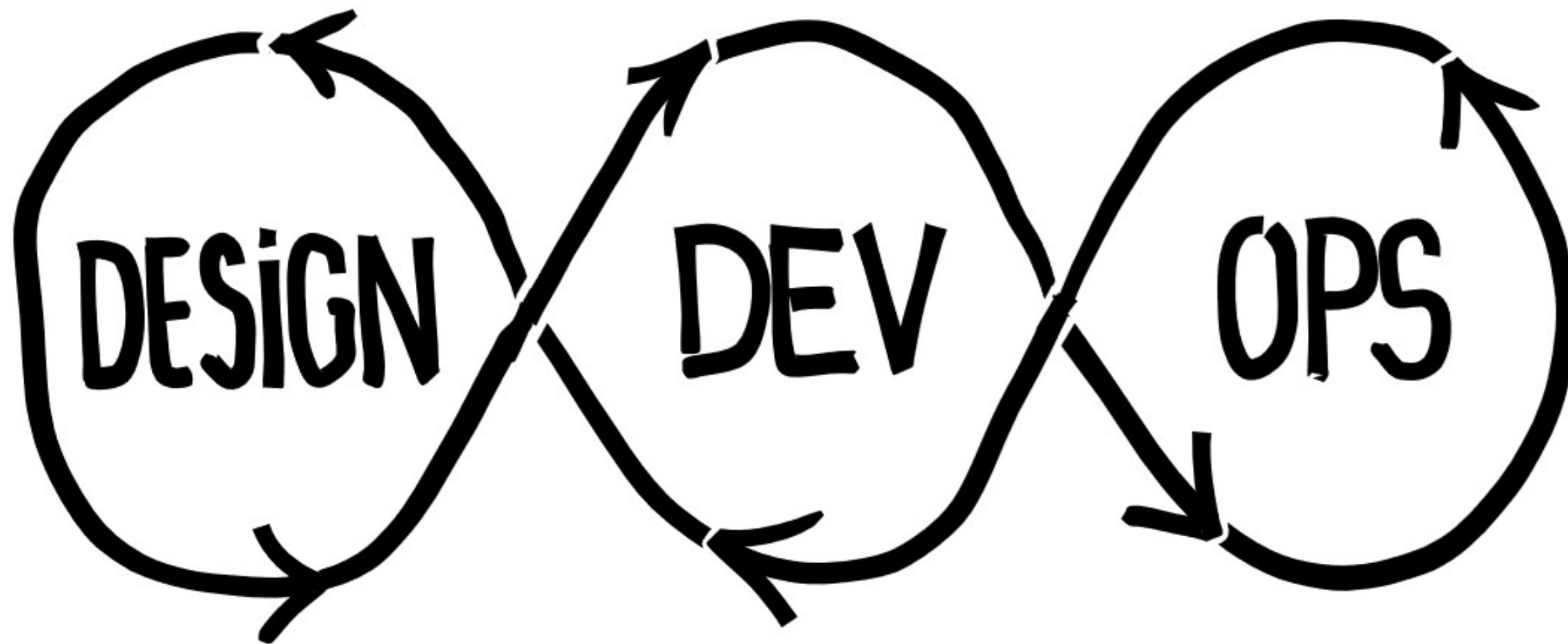
Effective model governance is the bedrock for minimizing risk to both an organization's bottom line and to its brand.

~ *DataRobot.com*



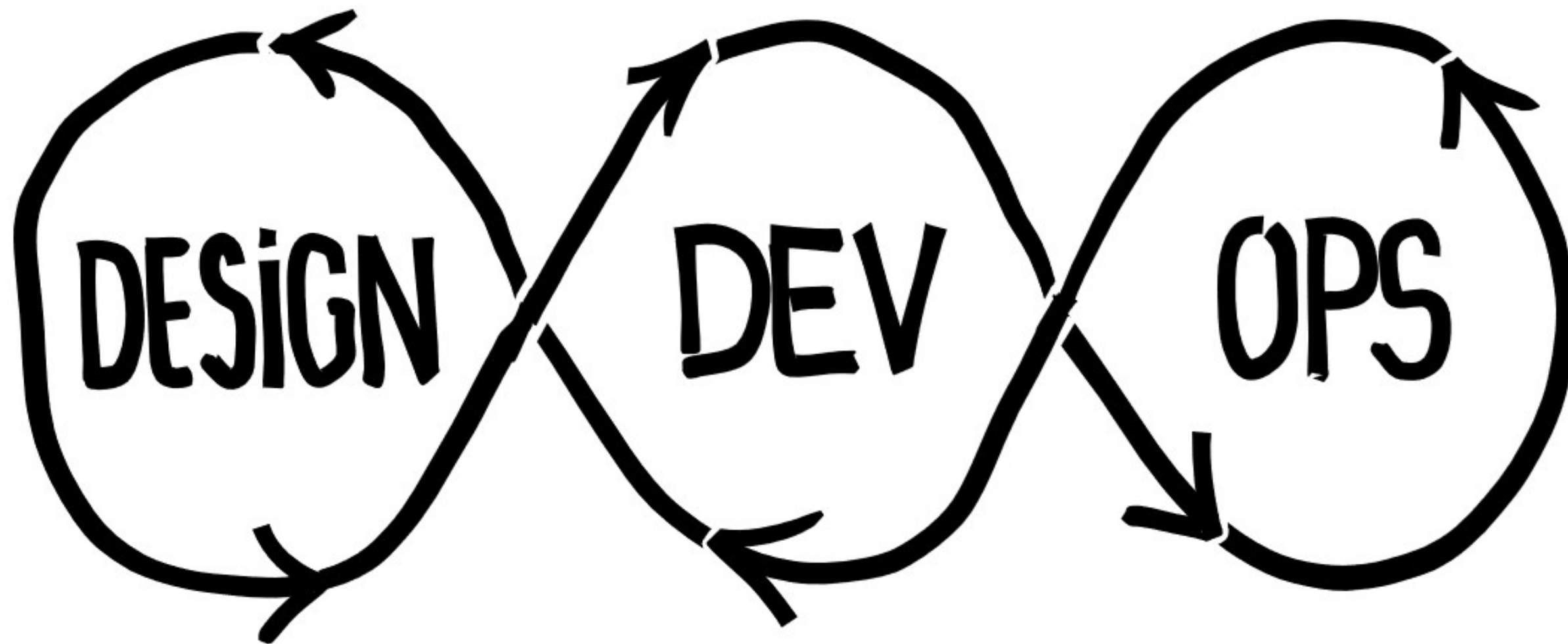


- Is it ethical to use ML?
- Private, sensitive data?
- Unethical model bias?
- ...



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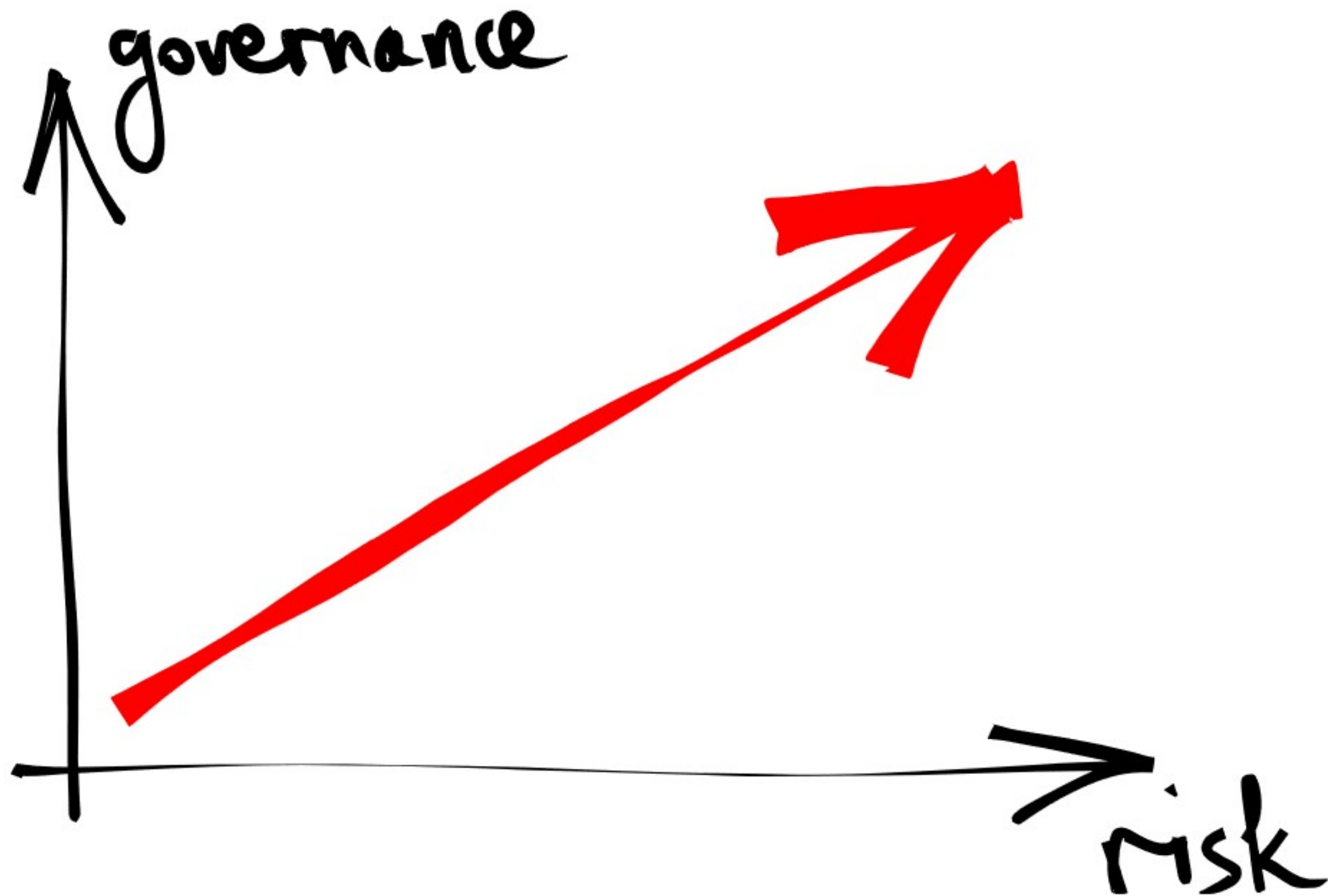
- Model selection docs
- Training data prep
- Quality assurance
- Versioning
- Reproducibility
- ...

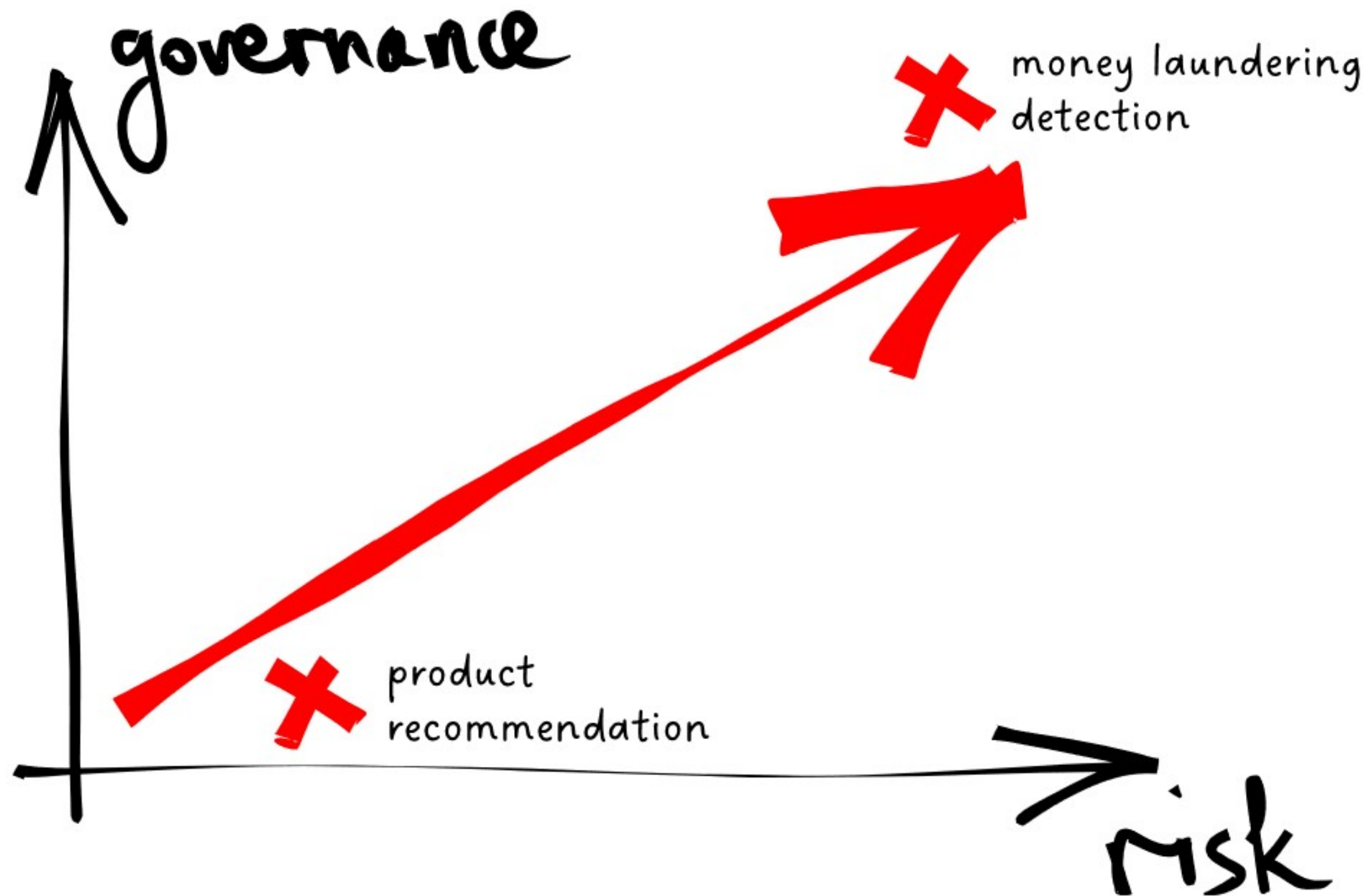


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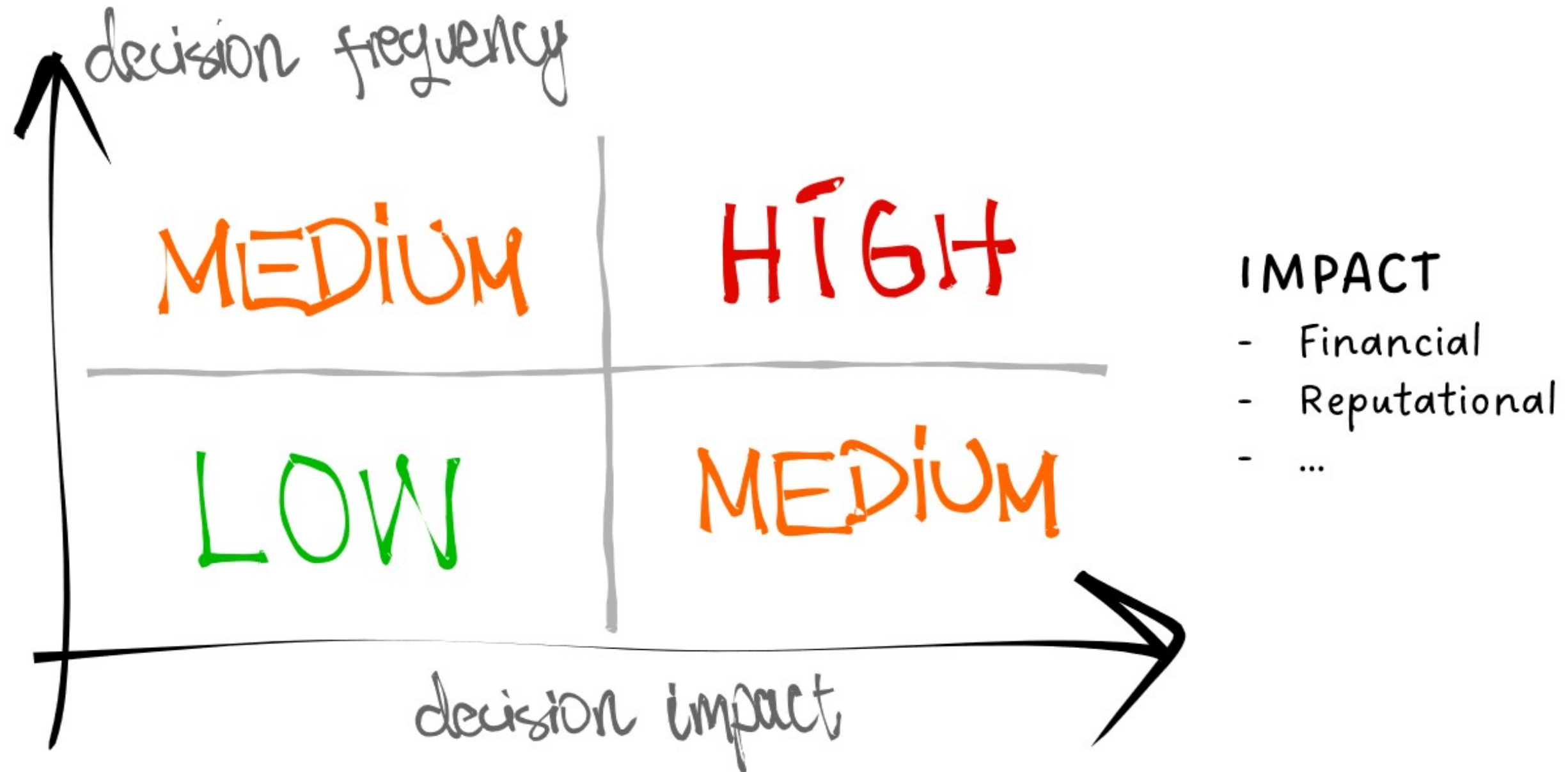
- Model selection docs
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- Versioning
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- ...

- Is API secure?
- Monitoring in place?
- Alerting?
- Failure handling?
- ...





Risk categories



Summary

- Governance means extra steps, but alternative is anarchy.
- "Launching many models fast" is not our goal, but generating business value.
- Reckless ML --> more damage than benefit.
- More models, more obvious the benefit of governance becomes.

Let's practice!

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Wrap up

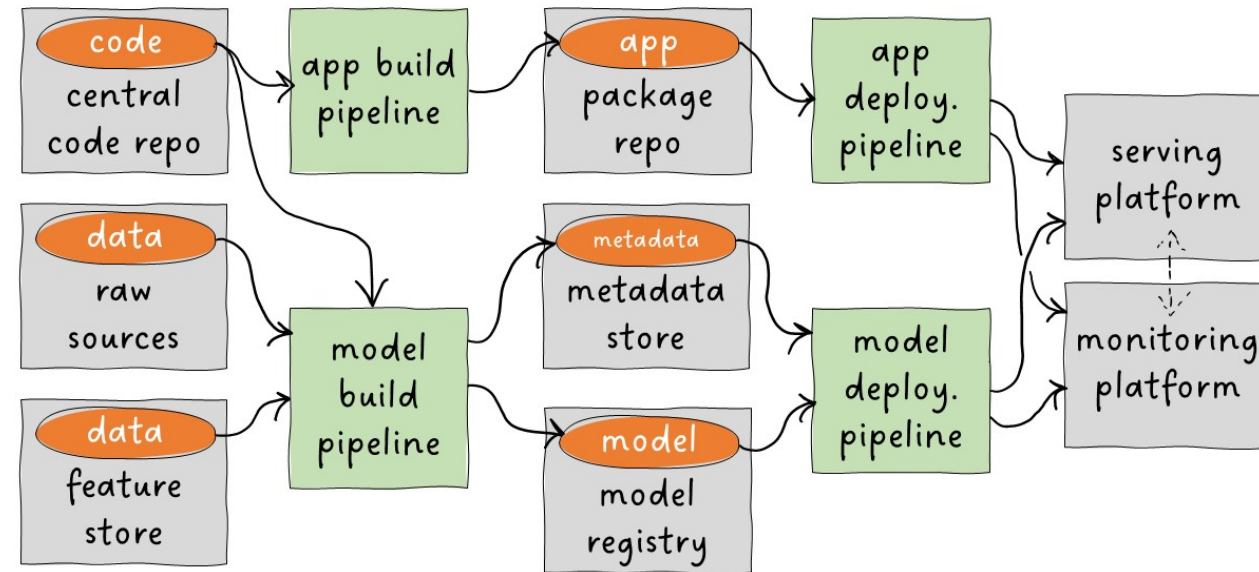
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Nemanja Radojkovic

Senior Machine Learning Engineer

Principles above all



#automation

#collaboration

#efficiency

#transparency

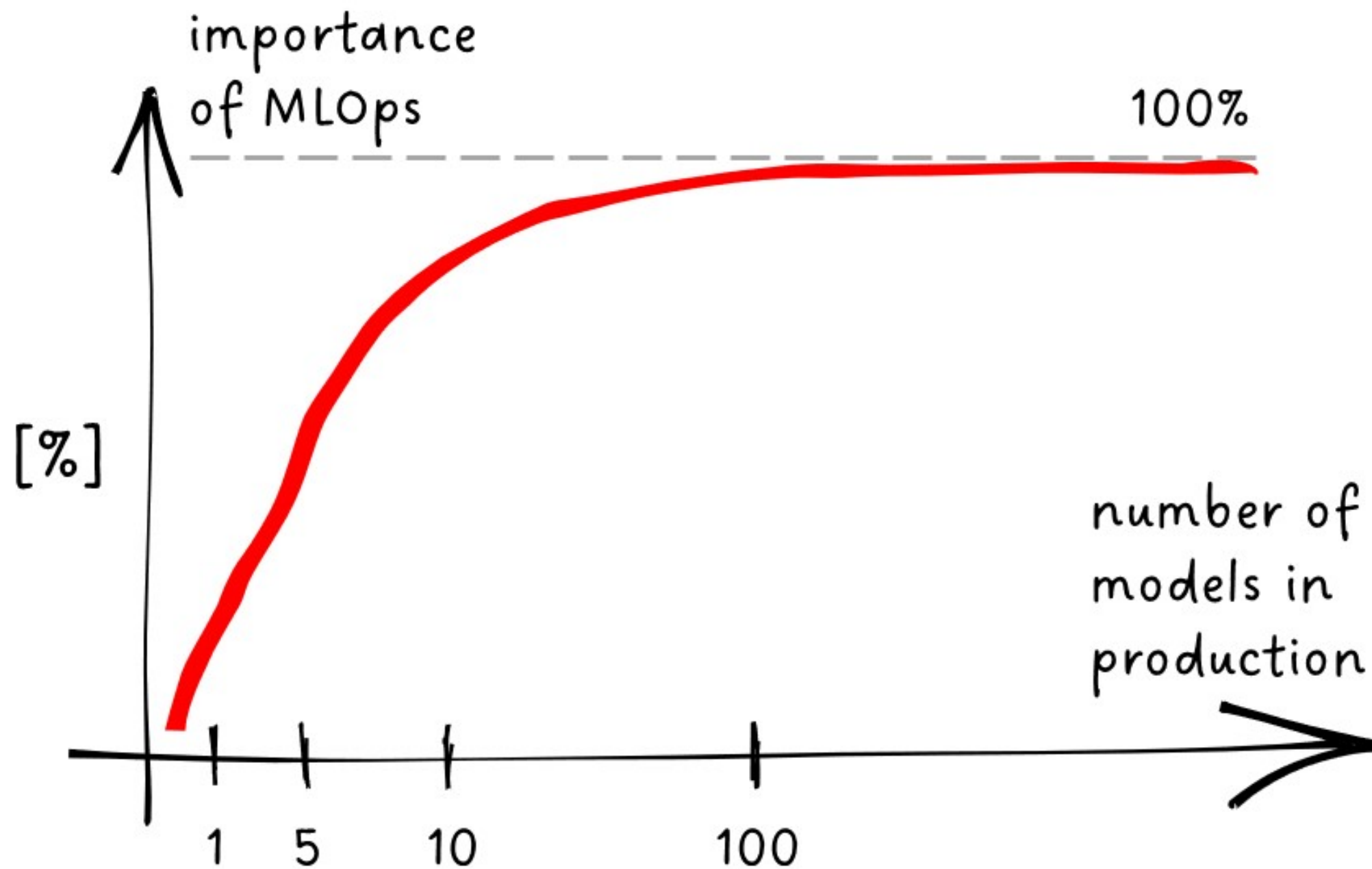
#user-satisfaction

mlflow™



great_expectations





Start small and grow



Your MLOps framework today



Your MLOps framework in five years

Thank you!

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